A Low-Cost and Low-Complexity Motion Sensor System Using IMU and Video-Based Tracking

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Abstract

This report presents MoveSync, a low-cost, browser-based motion analysis prototype developed for sports performance analysis. The system integrates two complementary sensing modalities, video-based pose estimation and inertial measurement unit (IMU) sensors, within an open-source, client-side web application requiring no downloads or server infrastructure. Video pose estimation is performed using MoveNet via TensorFlow.js, extracting two-dimensional skeletal keypoints from uploaded recordings. IMU sensor data is processed through a multi-stage pipeline comprising low-pass filtering, bias calibration, sensor fusion, gravity removal, and zero-velocity update (ZUPT)-based integration to derive motion metrics including peak acceleration, peak velocity, and total distance. A manual event-based synchronisation method aligns the two data streams, enabling coherent visualisation of the multimodal motion data. The application is structured around a project–session data model and comprises seven functional modules: Dashboard, Sport Presets, Upload, Library, Session Viewer, Compare Sessions and Processing Pipeline.

A central focus of this work is the investigation and improvement of the IMU sensor fusion pipeline. Three algorithms were implemented and evaluated against synthetic ground-truth datasets: an Expanded Madgwick filter, the Versatile Quaternion-based Filter (VQF), and an Error-State Extended Kalman Filter (EKF). The Expanded Madgwick filter performed unacceptably on rotation scenarios, accumulating yaw errors exceeding 1293° against a ground truth of 180°. VQF was selected as the optimal algorithm based on its superior performance on mixed translational-rotational motion, and a parameter sensitivity study identified $\tau_{\text{acc}} = 15\,\text{s}$ and $\tau_{\text{mag}} = 60\,\text{s}$ as practically optimal values. Physical validation experiments confirmed accurate yaw tracking (+5.2 % mean error across 90° rotation steps) but revealed substantial distance overestimation under low-acceleration conditions (+489 %), consistent with the known limitations of double integration from MEMS IMUs. These findings identify inertial distance estimation as the primary constraint on system accuracy and the main direction for future development. Despite these limitations, MoveSync demonstrates the feasibility of accessible, browser-based multimodal motion capture and establishes a functional foundation for further development.

Contributions

The development of the MoveSync system has been a collaborative effort, with overall contributions shared equally among group members. This includes the design and implementation of the web application codebase, the preparation and delivery of the group presentations, and the writing of all reports. All members attended weekly meetings with the project supervisor as well as additional group meetings when necessary, contributing equally during these.

Jack Hutton

Jack Hutton primarily led the IMU signal processing pipeline for this project. This involved a systematic investigation and comparison of three sensor fusion algorithms (Expanded Madgwick, VQF, and EKF), the design and generation of synthetic validation datasets, and the execution of physical validation experiments for orientation and mixed-motion tracking. He also implemented key stages of the processing pipeline within the web application, including bias calibration, acceleration gating, ZUPT and rotation gate detection, gravity removal, and velocity integration. He made multiple contributions to the web application development, primarily focusing in CM2017 on the Processing Pipeline and Session Viewer pages. Throughout the semester he contributed fixes across multiple features and components as problems arose. For this report, he was responsible for Sections 2.3, 2.4, 3.3–3.8, 4.2–4.4, 5.2–5.5, the Abstract and the Conclusion. He additionally contributed to the overall report through planning, editing, and proofreading.

Alexander Jonsson

Alexander Jonsson led the feature extraction. Specifically, Alexander decided which metrics were to be kept and which metrics were redundant or unnecessary. This also included reviewing how each metric is calculated, how each metric is implemented in the code, how the metrics are presented, and considering alternative approaches. All of this involved extensive literature reading to find the most suitable metrics and methods of choice. Furthermore, Alexander worked on debugging important issues, most notably the sensor fusion in the early part of the project, but also trying to improve metric accuracy by investigating the ZUPTs. For the report, Alexander was responsible for sections 2.5, 3.5, 5.6, and 5.7, partly responsible for sections 2.1 and 2.2, and also contributed to other sections by proofreading and giving suggestions.

Sahand Khodabandehloo

Sahand Khodabandehloo primarily contributed to the development of the web application structure and the underlying data architecture of the MoveSync system. His main contributions concerned the design of the project–session-based structure, including how video recordings and IMU data are organized, associated, stored, and handled within the application. He also contributed to the overall web application workflow, particularly the connection between project management, data upload, session inspection, processing, and session comparison. Within the report, Sahand was mainly responsible for sections 3.1, 3.2, 4.1, 5.1, and 5.8, and partly responsible for sections 2.1 and 2.2. In addition to these contributions, he contributed to the remaining sections through proofreading, editing, and suggestions aimed at improving the clarity, consistency, and academic quality of the report.

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1 Introduction

1.1 Motivation

In competitive sports, small margins often determine the difference between success and failure. Improvements in time, joint angles, force application, and movement coordination can have a substantial impact on athletic activities. For this reason, biomechanical analysis has become an increasingly important component of modern sports science, supporting evidence-based coaching, performance optimisation, and injury prevention (1). Yet, despite extensive scientific knowledge, many training decisions still rely heavily on visual assessment and subjective coaching experience rather than objective biomechanical data. Although expert observation remains valuable, it is inherently limited in precision, consistency, and reproducibility. This gap between scientific methods for motion analysis and their practical use in everyday training environments (1) creates a clear need for accessible tools that can provide quantitative movement analysis without requiring expensive laboratory equipment or highly specialised technical expertise.

Recent advances in wearable sensors, computer vision, and browser-based technologies have created new possibilities for low-cost motion capture systems. In particular, video-based pose estimation and wearable sensors offer complementary sources of biomechanical information: on the one hand, video-based pose estimation provides interpretable visual representations of body movement; on the other hand, wearable sensors capture acceleration and rotational motion (1). Integrating these technologies into a single browser-based application could enable practical and reproducible sports performance analysis in real-world settings.

1.2 Aim and objectives

This project aims to design a low-cost, low-complexity, and accessible browser-based motion-tracking prototype for sports performance analysis. The proposed system, *MoveSync*, is intended for athletes, coaches, and other users performing sports-related movements using video recordings, with optional support for IMU sensor data.

To achieve this aim, three objectives were defined to structure the development and evaluation of the project. These objectives are presented below:

1. To perform browser-based video pose estimation to extract two-dimensional skeletal keypoints from recorded sports movements.

2. To develop a synchronisation method capable of aligning video frames with IMU data despite their independent time bases.

3. To provide an intuitive interface for the visualisation of synchronised video and sensor data to support performance analysis.

2 Background

2.1 Motion Capture Technologies

Motion capture is the process of translating physical movement in 3D space into measurable data, enabling accurate reconstruction and analysis of motion (2). Current systems (e.g., optical, electromagnetic, and computer vision) differ in accuracy, portability, and practicality, requiring context-specific selection. Laboratory-based optical systems are typically restricted to controlled environments due to technical complexity and resource demands (3). In contrast, alternative systems such as wearable sensors and vision-based methods involve trade-offs in performance and deployment constraints (4).

In sports, motion capture is used to inform training decisions in dynamic environments (1, 4), as it supports technique assessment, performance monitoring, asymmetry detection, and efficiency evaluation to guide performance enhancement and injury prevention. However, not only is access to laboratory-grade systems limited but also their application in real-world settings remains challenging, creating demand for more accessible and practical solutions (4).

2.2 Video Pose Estimation

Video pose estimation infers human pose from RGB video frames by predicting a set of 2D body keypoints that can be visualised as a skeletal overlay (5). This project uses MoveNet via the TensorFlow.js pose-detection API, running directly in the browser. MoveNet is a pose detector that outputs keypoints along with confidence scores for each detected joint, enabling reliable tracking of body position across video frames. It is designed for fast, robust, and temporally stable performance, thereby reducing jitter and producing smoother motion trajectories suitable for video analysis (6). Of its two variants, Lightning and Thunder, the Thunder model is used in this project because it prioritizes higher accuracy and greater stability over raw speed (6), making it more appropriate for offline processing of uploaded videos where precision is more important than real-time performance.

2.3 Inertial Measurement Units

An Inertial Measurement Unit (IMU) is a compact electronic device that quantifies motion through the combined output of multiple embedded sensors: a three-axis gyroscope, a three-axis accelerometer, and, in many higher-capability devices, a three-axis magnetometer (7). Together, these yield up to nine degrees of freedom (DOF) of raw kinematic data from which the full three-dimensional orientation of the sensor can be estimated (8).

The most common type of IMU devices are micro-electro-mechanical systems (MEMS), which can range in size and complexity. In MEMS gyroscopes, angular velocity is measured by exploiting the Coriolis effect. An internal proof mass is driven to oscillate at its resonant frequency along a primary axis. As the device rotates, the Coriolis force (the apparent lateral deflection experienced by any moving body within a rotating reference frame) displaces the proof mass perpendicular to both its direction of oscillation and the rotation axis. This deflection is proportional to the angular velocity and is detected capacitively, producing an electrical signal that is converted to an angular rate (9, 10). This principle is illustrated in Figure 1. The accelerometer operates on a similar principle with a suspended proof mass that is displaced relative to its housing by any applied force, including both inertial acceleration and gravity. This displacement is also detected capacitively and converted to acceleration (7). The magnetometer measures the local magnetic flux density as a reference for absolute heading.

Most IMUs used in human motion analysis are MEMS devices as they typically have a small form factor, low power consumption, and low cost (8). These properties make them more reliable and usable in the field. However, MEMS sensors are subject to well-known error sources,

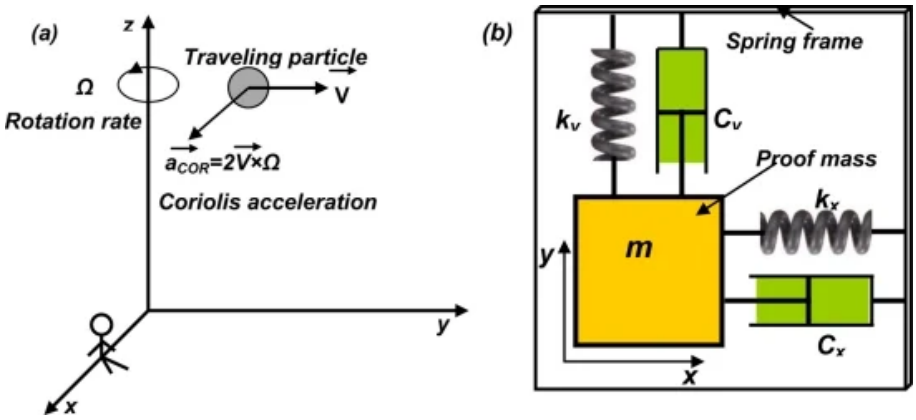


Figure 1: The Coriolis effect (left) and MEMS gyroscope proof-mass mechanism (right), illustrating the physical basis of angular rate measurement in MEMS gyroscopes (11).

including thermal drift, scale-factor nonlinearity, broadband noise, and static bias (9). Because orientation estimation and kinematic metric extraction both rely on numerical integration of these signals, errors accumulate over time and must be addressed through signal processing before reliable results can be obtained.

2.4 Signal Processing

A multi-stage processing pipeline can be applied to the raw IMU output to correct measurement errors and extract reliable kinematic quantities. The key stages are digital filtering, bias calibration, sensor fusion for orientation estimation, gravity removal, and drift bounding for velocity integration.

2.4.1 Filtering

A low-pass filter is applied to the raw signals prior to integration to attenuate broadband noise (12). Because integration accumulates errors over time, small residual noise components can produce growing drift in estimated velocity and displacement (13). The cutoff frequency must be chosen to suppress noise while retaining genuine motion content. Human motion frequency content is generally reported below 5 Hz for common activities (14), with upper bounds of up to 12 Hz reported when body segment accelerations are considered (15). Dynamic sport-specific movements involving rapid impacts or high-speed limb motion may therefore require filter cutoff frequencies toward the upper end of this range.

2.4.2 Bias Calibration

IMU devices are affected by systematic sensor bias that introduces persistent errors into their raw measurements (9). Higher quality devices often have lower biases but this can come with a significantly higher cost. These biases, if uncorrected, can degrade the accuracy of motion and orientation estimates. They can arise from imperfections in the fabrication of the device or from environmental and electromagnetic influences.

Accelerometers can exhibit bias that typically appears as a constant offset in the measured acceleration. This can have significant impacts when acceleration is integrated to estimate velocity and position, where small offsets can lead to significant drift over time. Accelerometer bias also affects gravity estimations, which are important for orientation estimation in sensor fusion algorithms (9).

Gyroscopes are particularly sensitive to bias as well, often having a small constant offset (static bias) that can lead to an error accumulating in the estimated orientation. This error propagates into all the derived quantities that depend on orientation, including processed accelerations, velocities, and displacements (9).

Magnetometers are subject to a hard-iron bias, caused by the magnetism from nearby ferromagnetic materials or components within the device itself. This appears as a constant offset in the measured magnetic field vector, shifting the magnetic field uniformly and distorting heading estimates. There are also soft-iron effects which can alter the shape of the measured magnetic field (9).

Commonly, these biases will be measured and removed through a calibration process prior to sensor fusion (10). Accurate bias estimation is essential for reducing drift and ensuring long-term accuracy in IMU systems.

2.4.3 Sensor Fusion

Each of the sensors within an IMU device has complementary error profiles. The gyroscope offers high bandwidth and low short-term noise but drifts through bias and integration error, while the accelerometer and magnetometer provide stable long-term reference vectors for gravity and magnetic north, respectively, but are susceptible to noise and short-term disturbances (16). Sensor fusion combines these signals to balance their respective strengths and weaknesses in estimating orientation. This combination typically assumes that the accelerometer measures gravity, an assumption that does not always hold under dynamic motion.

Accelerometer-assisted orientation estimation is limited by the fact that the accelerometer measures *specific force*, i.e. the combined effect of gravity and linear acceleration. These components are inseparable in the measured signal, so a tilted static sensor and a translating sensor can produce identical outputs, as illustrated in Figure 2 (17). As a result, treating the accelerometer as a gravity reference is only valid under quasi-static conditions. During dynamic motion (e.g. sprinting or jumping), this assumption breaks down, corrupting the orientation estimate and all derived kinematic quantities. This motivates sensor fusion methods that adaptively weight or reject accelerometer data during motion.

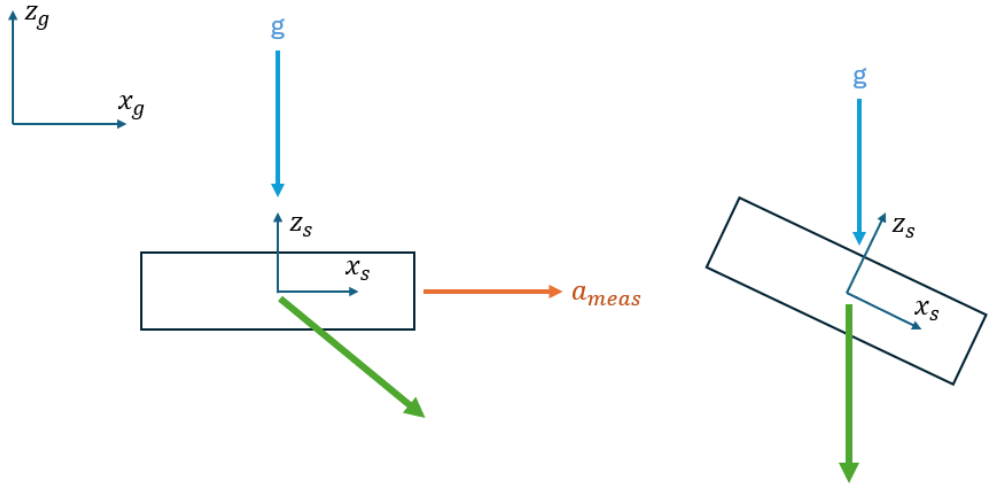


Figure 2: The specific force ambiguity demonstrated. A sensor undergoing horizontal linear acceleration (left) experiences a specific force vector (green) identical in the sensor frame to a stationary sensor that is tilted (right).

2.4.4 Orientation Representation

Orientation in three-dimensional space can be represented using Euler angles, rotation matrices, or quaternions. Euler angles describe orientation as three sequential rotations (e.g. roll, pitch, yaw), but suffer from gimbal lock, a singularity that occurs when two rotation axes align. For example, at a pitch of $\pm 90°$, the roll and yaw axes coincide, resulting in a loss of one degree of freedom. This is illustrated in Figure 3. Rotation matrices avoid this issue but require nine elements subject to orthonormality constraints. Quaternions provide a compact and numerically stable alternative.

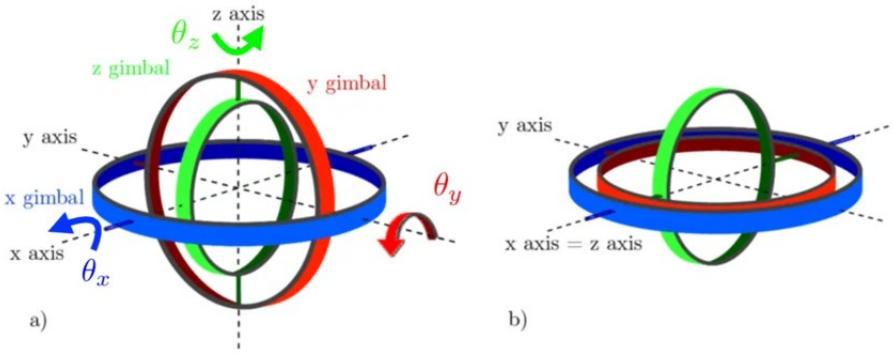


Figure 3: Gimbal lock in Euler angle representation. Under normal conditions (a) the three gimbal axes are independent. When pitch reaches 90° (b) the x and z axes become collinear, collapsing the system to two effective degrees of freedom (18).

A rotation can be expressed as an axis–angle pair: a unit vector $\hat{\mathbf{n}} = (n_x, n_y, n_z)$ and a rotation angle θ. A unit quaternion encodes this as

$$q = \cos\frac{\theta}{2} + \sin\frac{\theta}{2}(n_x\mathbf{i} + n_y\mathbf{j} + n_z\mathbf{k})$$

with the constraint

$$w^2 + x^2 + y^2 + z^2 = 1.$$

Although four parameters are used, the unit-norm constraint leaves three effective degrees of freedom. This representation is free of singularities and computationally efficient, making it standard for IMU-based orientation estimation (19). This is illustrated in Figure 4.

2.4.5 Madgwick Filter

The Madgwick filter is a computationally efficient orientation estimator based on quaternion kinematics and gradient descent (20). Gyroscope measurements are integrated to propagate orientation, introducing drift due to bias. Accelerometer and magnetometer measurements are incorporated to prevent this by minimising the error between estimated and measured reference vectors. A gradient descent step provides a correction term that is combined with the gyroscope prediction.

A single gain parameter β controls the balance between integration and correction. Larger values increase responsiveness but also the sensitivity to disturbances, while smaller values reduce noise influence at the cost of increased drift. This fixed-gain structure can limit performance during highly dynamic motion.

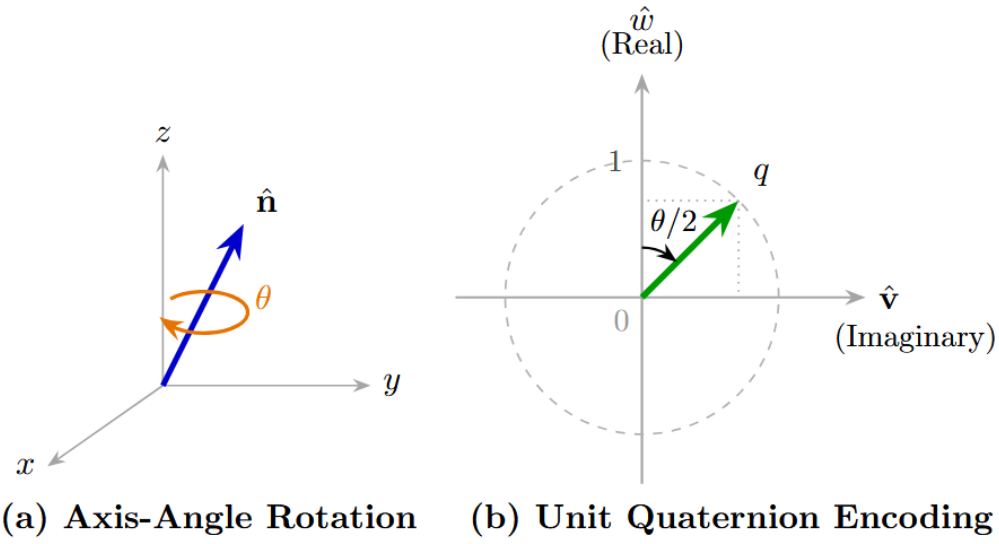


Figure 4: Axis–angle representation of a rotation and its unit quaternion encoding. (a) A rotation is defined by a unit axis $\hat{\mathbf{n}}$ and angle θ. (b) The unit quaternion maps this onto a 2D constraint circle using the half-angle $\theta/2$, where the scalar part $\hat{w} = \cos(\theta/2)$ and imaginary vector part $\hat{\mathbf{v}} = \sin(\theta/2)$ satisfy the unit norm constraint $\hat{w}^2 + \hat{\mathbf{v}}^2 = 1$, avoiding the singularities associated with Euler angle representations (19).

A common limitation of the standard Madgwick filter is its sensitivity to accelerometer corruption during dynamic motion, where linear acceleration violates the assumption that the measured signal corresponds to gravity alone. A simple extension to address this is acceleration gating, in which the accelerometer correction step is disabled whenever the magnitude of the measured acceleration deviates from the expected $1\,g$ gravity norm. During these intervals, orientation updates rely solely on gyroscope integration, which avoids contamination from linear acceleration but allows drift to accumulate. Once the acceleration returns within the valid range, the correction step is reactivated. This modification preserves the computational efficiency of the original filter while improving robustness in highly dynamic conditions.

2.4.6 Versatile Quaternion-based Filter

The Versatile Quaternion-based Filter (VQF) was introduced by Laidig and Seel in 2023 and is designed specifically for wearable IMU applications (16). It separates inclination estimation (roll and pitch via gravity) from heading estimation (which additionally uses the magnetometer). Unlike fixed-gain approaches, VQF incorporates adaptive mechanisms for handling sensor disturbances.

Acceleration rejection is a core component of the inclination estimation stage. Rather than applying a binary gate, VQF continuously adjusts the influence of the accelerometer based on deviation from the $1\,g$ gravity norm, enabling smooth down-weighting during dynamic motion. In addition, the filter performs online estimation of gyroscope bias within each stage, improving long-term stability without external calibration.

In published benchmarks on standard IMU evaluation datasets, VQF outperformed the Madgwick filter across a range of conditions, with the largest improvements observed during sustained dynamic acceleration (16).

2.4.7 Error-State Extended Kalman Filter

The Extended Kalman Filter (EKF) is a recursive Bayesian estimation framework that linearises a nonlinear system model around the current estimate and propagates both the state and its uncertainty covariance forward in time (21). For orientation estimation, the error-state (multiplicative) formulation is commonly used, as it avoids numerical issues associated with directly parameterising quaternion uncertainty (22).

In this formulation, the quaternion is propagated using gyroscope integration, while a separate linearised error state is maintained to represent small orientation deviations and sensor biases. When accelerometer or magnetometer measurements are available, they are used to correct this error state via the Kalman gain. Because the filter maintains an explicit covariance, measurement uncertainty can be adjusted dynamically, allowing reduced reliance on accelerometer data during periods of high dynamic acceleration. This principled treatment of uncertainty makes the error-state EKF the most theoretically rigorous of the three approaches mentioned here, at the cost of increased implementation complexity.

2.4.8 Gravity Removal

Raw accelerometer data contains contributions from both linear acceleration and gravity. Since gravity is fixed in the world frame but projects onto all sensor axes depending on orientation, direct removal requires knowing the current attitude (10). Using the orientation quaternion from the fusion stage, the world-frame gravity vector is rotated into the sensor body frame via the conjugate quaternion rotation and subtracted from the raw reading. The residual linear acceleration can then be rotated back into the world frame for consistent downstream interpretation. Orientation accuracy and gravity removal accuracy are directly coupled such that any orientation error leaves a residual gravity component in the output, introducing a sustained bias into all integrated quantities (10, 23).

2.4.9 Zero-Velocity Updates

Velocity estimation via integration is prone to drift due to accumulated errors. Zero-velocity updates (ZUPTs) mitigate this by identifying stationary intervals and enforcing a zero-velocity constraint (24). These stationary intervals can be detected using thresholds on gyroscope and accelerometer magnitudes. During these periods, velocity is reset to zero, bounding drift over time.

In periodic motion, cadence can provide an additional temporal constraint that can be used to structure the estimation process. Rather than relying solely on detected stationary phases, cadence can be used to impose regular re-initialisation or correction points within the motion cycle, either as a supplement to ZUPT events or as a substitute when zero-velocity conditions are weak, absent, or unreliable (25).

A limitation of standard ZUPT detection is that it relies on low gyroscope magnitude as a stillness criterion, which excludes pure rotational motion. During rotation with no translational component, the accelerometer continues to read $\approx 1\,g$ (gravity only), yet velocity integration proceeds and any residual linear acceleration from orientation error accumulates into phantom displacement. A rotation gate addresses this by applying the zero-velocity constraint under the complementary condition: gyroscope magnitude is high (indicating active rotation) while specific-force magnitude remains close to $1\,g$ (confirming the absence of translation). Under this condition, suppressing velocity prevents gravity-removal residuals from integrating into spurious distance during rotational phases (24).

2.5 Sports Analysis Metrics

Quantitative metrics play a central role in sports analysis by providing objective measures of athlete performance, thereby reducing reliance on subjective judgment. This section presents key metrics used in sport science, grouped into acceleration-based, velocity-distance, and angular velocity metrics.

2.5.1 Acceleration metrics

Acceleration-based metrics are widely used in IMU-based sports analysis as they provide direct information about movement intensity and explosiveness (26). In raw IMU measurements, acceleration includes both dynamic body motion and the constant contribution of gravity. Thus, by removing gravity from the IMU data, the acceleration induced by user movement can be isolated (27). Furthermore, acceleration-based metrics have been used to monitor external load and are associated with injury risk in some sporting contexts (28).

There are multiple acceleration metrics that are useful. For example, peak acceleration, the highest measured acceleration during a session, has been used to assess injury risk during high-impact movements such as landings or sprint foot strikes (29). Continuous measures of acceleration load, such as acceleration Root Mean Square (RMS), are also useful in sports science as they reflect the overall magnitude and intensity of movement over time, rather than isolated peak or threshold-based events (26). Finally, peak jerk, the maximum rate of change of acceleration, reflects how abruptly acceleration varies during movement. High jerk values are associated with sudden, less controlled movements and may increase injury risk, whereas lower values indicate smoother motion. For example, skilled golfers demonstrate lower peak jerk in lower-body joints compared to less skilled players (30).

2.5.2 Kinematic metrics

Kinematic metrics describe the overall movement output of an athlete, typically capturing how far and how fast the body or body segments move during an activity. In IMU-based sports analysis, these measures are commonly derived by integrating acceleration data over time to estimate velocity and displacement. However, this process is sensitive to measurement noise and small biases in the acceleration signal, which can accumulate over time during integration. To mitigate this issue, ZUPT techniques are sometimes applied during periods of known or assumed stationary motion to help constrain drift in the estimated velocity (31).

A prevalent kinematic metric is peak speed which is the highest speed reached during a session. Higher peak speeds have been shown to correlate with match outcomes in soccer (32). Another useful metric is mean velocity. This metric has been shown to predict relative load in the bench press throw more accurately and linearly than peak or mean propulsive velocity (33). Additionally, a good metric to quantify overall workload is the cumulative distance the sensor has moved, which helps coaches monitor fatigue and reduce injury risk (34).

2.5.3 Rotational metrics

Orientation-related metrics describe the rotational movement and spatial orientation of the body or body segments during athletic activities. These measures are commonly derived using sensor fusion techniques (35). These types metrics are widely used in sports analysis to assess movement technique, stability, and coordination.

One group of orientation-based metrics is angular velocity metrics, which measures the speed at which something rotates. Peak angular velocity is important for instance in golf, where higher peak angular velocities of the shoulders and pelvis are linked to better performance (36), and in javelin throwing, where greater trunk and shoulder angular velocities are associated with higher

release speeds and longer throws (37). As with acceleration metrics, RMS can also be used to summarise angular velocity signals, providing an estimate of overall magnitude and movement intensity over time.

Another group is range-of-motion (ROM) metrics. These quantify the range of motion of a body segment or joint over a movement, typically defined as the difference between maximum and minimum joint angles. ROM is therefore a useful indicator of joint mobility and movement capacity, and is widely used in sports science to assess movement quality and to support both performance optimisation and injury risk reduction (38).

2.5.4 Temporal performance metrics

Temporal performance metrics describe the timing structure and participation characteristics of athletic activity. These metrics capture how frequently movements occur and how long athletes are active during a given task or session. In IMU-based sports analysis, they are typically derived from event detection algorithms that identify discrete movement cycles or activity periods (39).

A common temporal metric is cadence, which describes the frequency of repeated movement cycles, such as steps or repetitions per unit time. Higher cadence has been shown to influence running biomechanics and energetic cost (40). Another useful metric is repetition count, which quantifies the total number of discrete movement cycles performed during a session.

In addition, mean repetition time describes the average duration of each movement cycle, providing insight into movement tempo and execution strategy. This metric can be useful for comparing technique between athletes or monitoring fatigue-related changes in movement speed. Finally, active time represents the total duration during which meaningful movement is detected above a defined threshold. This measure can be used to quantify overall participation or exposure.

3 Methods

3.1 System Architecture Design Principles

The *MoveSync* system was developed as a modular web-based client-side platform using HyperText Markup Language (HTML), Cascading Style Sheets (CSS), and JavaScript (JS). These technologies were selected because they constitute the fundamental building blocks of interactive web applications and enable the system to operate directly within a web browser without requiring additional software installation.

HTML was used to define the structural organization and content of the user interface, including input fields, control buttons, data sections, and visualization containers. CSS was used to specify the visual presentation of the application, including layout, spacing, styling, and responsive design, thereby contributing to a clear and user-oriented interface. JavaScript was used to implement the interactive and dynamic functionality of the system, including user interaction, data handling, server communication, motion-data processing, and real-time updates.

Together, these technologies provided a structured framework for managing, processing, and visualizing multimodal motion data. The modular architecture of the system enables individual components to be modified, replaced, or extended independently, thereby improving maintainability, scalability, testing, and debugging.

3.1.1 Project–Session–Based Data Model

Because the system handles multiple datasets in different formats, data are organized in a hierarchical *project–session* structure (see Figure 5). A *session* consists of one video recording and its associated IMU data files, forming the basic unit for storage, processing, and analysis. A *project* contains multiple sessions, allowing related datasets from the same movement, experiment, or analysis context to be grouped.

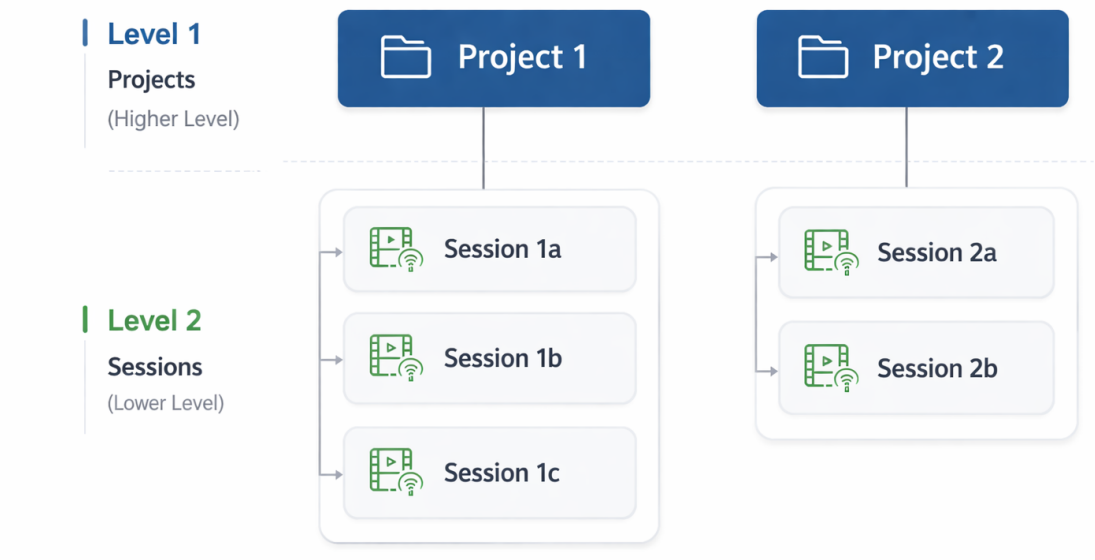


Figure 5: Illustration of the hierarchical *project–session* structure. A *project* forms the top-level container, while multiple *sessions* are nested beneath it. Each session corresponds to one complete data import and is treated as an individual analytical unit.

3.1.2 Data Stream Processing

Video-based pose estimation data and IMU data were treated as separate data streams, since they were generated from different sensing modalities and may contain different sampling rates, timestamp formats, and recording start times. Each stream was therefore preprocessed independently through parsing, formatting, and validation before being used for combined analysis. During preprocessing, the relevant timestamp information was extracted and converted into a consistent temporal representation to enable comparison between the data streams.

3.1.3 Temporal Synchronization

Temporal synchronization was performed in the Session Viewer by calculating an offset time between the data streams. The offset time represents the temporal difference between the time reference of the video and the corresponding time reference of the IMU data. It was computed as the difference between the corresponding reference timestamps of the streams (e.g., the first valid timestamp or another defined synchronization point). In this application, the video time was used as the reference, and the IMU data was shifted relative to the video time. The offset time was calculated as

$$t_{\text{offset}} = t_{\text{video}} - t_{\text{IMU}}$$

where t_{video} represents the reference time of the video data and t_{IMU} represents the corresponding reference time of the IMU data. The calculated offset time was then used to synchronize the IMU timestamps with the video timeline:

$$t_{\text{sync}} = t_{\text{IMU}} + t_{\text{offset}}$$

This procedure maps the different data streams onto a common session time axis. As a result, pose estimation data and IMU measurements can be visualized and analyzed according to the same temporal reference, despite differences in their original recording start times or timestamp formats.

In the Session Viewer, the computed offset is used during playback and visualization to determine which samples from each stream correspond to the same point in time. This enables synchronized inspection of movement data across modalities while preserving the original data structure of each stream. The synchronization step therefore corrects for a constant temporal displacement between streams, but does not alter the underlying sampling frequency or independently resample the raw data. This design maintains flexibility in handling heterogeneous data sources and supports modular processing, visualization, and debugging.

3.2 Web Application Design and Implementation

The MoveSync web application was developed as a centralized platform for managing, visualizing, and analyzing motion-tracking sessions. An iterative development process was used to refine the user interface, improve functionality, and integrate new features while maintaining a consistent application structure. The application was organized into several modules that together support the full motion-analysis workflow.

The *Dashboard* module provides an overview of recent sessions, stored projects, and general system status. The *Sport Presets* module allows predefined analysis settings to be created for different movement tasks, including selected metrics, timestamp categories, and standardized analysis windows. These presets reduce repeated manual configuration and support consistent evaluation across sessions. The *Upload* module enables import of video recordings and IMU data in CSV format and their association with a project. During upload, the session structure is initialized for subsequent processing and analysis. The *Library* module functions as a

repository for stored projects (containing sessions), allowing previously imported datasets to be retrieved, edited, or exported for further analysis. The *Session Viewer* module was developed for synchronized inspection of video and IMU-derived motion signals. It provides tools for time synchronization, playback control, visualization of sensor signals, and extraction of selected performance metrics. The *Compare Sessions* module enables two sessions to be displayed side by side, supporting visual and metric-based comparison of movement performance across recordings. Finally, the *Processing Pipeline* module allows the configuration of preprocessing and sensor-fusion parameters, including filtering, calibration, and zero-velocity update settings. This makes it possible to adjust the signal-processing workflow before the data are used for visualization and analysis.

The *Tutorials* module was developed to support user onboarding and provide guidance on how to use the web application. Unlike the main analysis modules, this module is intended as an instructional resource rather than a tool for processing or evaluating motion data. It presents an overview of the application workflow and includes step-by-step material covering project setup, preset configuration, data upload, session viewing, session comparison, and advanced analysis. This integrated guidance helps users understand the purpose of each part of the system and promotes consistent use of the application without relying solely on external documentation.

3.3 IMU Pre-Processing and Calibration

Prior to executing the sensor fusion algorithms, raw data in the form of a .csv file undergoes automated client-side normalization. The sensor sample rate is dynamically estimated by taking the median of the inter-sample time differences, and gyroscope units (radians vs. degrees) are verified using a 75th-percentile magnitude threshold. As an initial bias removal step, the per-axis median of the magnetometer signal is calculated and subtracted across the entire data stream before the signals are passed to the filters, neutralising hard-iron bias and protecting heading estimation from localised magnetic distortion.

3.4 Sensor Fusion Algorithm Configurations

In the process of investigating different methods of sensor fusion, three distinct fusion pipelines were implemented and evaluated. All three share a common kinematic back-end, undergoing the same overall process in this system. In each pipeline raw signals are first low-pass filtered, gyroscope bias is estimated from an initial static window and accounted for, orientation is computed by the selected algorithm, the gravity vector is rotated and subtracted from the accelerometer to obtain linear acceleration, and velocity is recovered by forward-Euler integration with zero-velocity updates (ZUPTs).

Two distinct ZUPT conditions are applied. A *stationary* ZUPT fires when the gyroscope magnitude falls below $8\,^{\circ}\mathrm{s}^{-1}$ and the specific-force magnitude lies within $[0.80, 1.04]\,g$. Once these conditions have been satisfied for at least eight consecutive samples the velocity is reset to zero. Conversely, a *rotation gate* is initiated when the gyroscope magnitude exceeds $30\,^{\circ}\mathrm{s}^{-1}$ and the specific-force magnitude is within $\pm 0.05\,g$ of $1\,g$. These thresholds were selected empirically to reliably identify stationary and rotational phases without falsely triggering during genuine motion.

3.4.1 Expanded Madgwick Filter

In previous work completed on this project the default sensor fusion filter was the Madgwick filter. This has some distinct problems and has been subsequently modified with specific additions to aid its accuracy. The Expanded Madgwick filter is the name given here to this modified version, extending the gradient-descent quaternion filter described in Section 2.4.5 with gyroscope

bias calibration and acceleration gating. The reasoning behind these additions is discussed in Section 5.2.1.

First, gyroscope bias is estimated by averaging the raw gyroscope signal over the initial static window (acceleration within $0.05\,g$ of $1\,g$, gyroscope below $2\,^{\circ}\mathrm{s}^{-1}$, sustained for at least 1.5 seconds) and subtracted from every subsequent sample before it enters the filter. Second, the accelerometer correction step is gated. As such it is disabled whenever the sensed specific-force magnitude departs from $1\,g$ by more than $0.1\,g$, at which point the filter propagates orientation by gyroscope integration alone until the magnitude returns within range.

In addition, a drift correction step was applied to the Expanded Madgwick filter only. After velocity integration, the residual velocity at the last moving sample before each ZUPT point is recorded. This value represents the accumulated phantom velocity arising from orientation error corrupting gravity removal. It is linearly subtracted across the full segment between ZUPT pairs, retroactively removing drift from the integrated velocity signal. This correction was not applied to VQF or EKF.

3.4.2 Versatile Quaternion-based Filter (VQF)

The VQF pipeline, as described in Section 2.4.6, is configured with two time constants that control the rate of tilt and heading correction. τ_{acc} governs tilt correction via the per-step gain $k_{\mathrm{acc}} = \Delta t/(\tau_{\mathrm{acc}} + \Delta t)$, and τ_{mag} governs heading correction via the analogous gain $k_{\mathrm{mag}} = \Delta t/(\tau_{\mathrm{mag}} + \Delta t)$. The default values used in the algorithm comparison are $\tau_{\mathrm{acc}} = 5\,\mathrm{s}$ and $\tau_{\mathrm{mag}} = 9\,\mathrm{s}$.

3.4.3 Error-State Extended Kalman Filter

The error-state EKF described in Section 2.4.7 is implemented with a binary measurement gate. As such accelerometer and magnetometer updates are accepted only when the sensed specific force lies within $\pm 0.2\,g$ of $1\,g$, with the Kalman gain set to zero outside this window so that orientation propagates by gyroscope integration alone.

3.5 Sports Metrics Calculation

The metrics that were included in this project are: Peak acceleration, acceleration RMS, peak jerk, peak speed, mean speed, total distance, peak angular velocity, angular velocity RMS, pitch, roll and yaw range-of-motion, cadence, repetition count, average repetition duration and active time.

Sensor fusion was used to estimate orientation and remove the gravity component from raw accelerometer data, which follows established IMU orientation estimations (41, 42, 43). Each accelerometer axis was then passed through a first-order IIR low-pass filter. For peak acceleration and RMS acceleration, the magnitude was computed using the L2 norm, and standard formulas were used for both RMS and peak extraction. Peak jerk was calculated by first differentiating each axis using a central-difference method, then combining the results into a jerk magnitude, following the procedure outlined by Simonsen et al.(44).

Velocity was estimated by trapezoidal integration of the filtered acceleration signal. To reduce drift over time, zero-velocity updates (ZUPTs) were applied during detected stationary periods. Total distance was then computed by integrating velocity. As with acceleration, only the magnitude of the velocity vector was used, again computed via the L2 norm.

Angular velocity metrics were derived from the L2 norm of the raw gyroscope signal without filtering, in order to preserve peak rotational events. Both peak angular velocity and RMS angular velocity were computed directly from this unfiltered signal.

Angular velocity metrics were computed from the L2 norm of the raw, unfiltered gyroscope

signal. Both peak angular velocity and angular velocity RMS were derived without filtering to preserve signal characteristics and avoid attenuation of peak rotational events.

Orientation angles were obtained from quaternion estimates produced by the sensor fusion filter, using a ZYX Euler angle decomposition. Roll, pitch, and yaw ranges were calculated using phase-unwrapped signals to avoid artificial discontinuities at the $\pm 180^\circ$ boundary.

Cadence, repetition count, and mean repetition time were estimated using peak detection on the smoothed accelerometer magnitude. A dynamic threshold of the form $\mu + k\sigma$ was used, where $k = 0.4$, μ is the mean of the signal, and σ is the standard deviation. This is a commonly used adaptive thresholding approach in wearable signal processing (45), with k typically tuned based on preprocessing and activity type (46). In this work, a slightly lower value of k was chosen due to prior smoothing and additional temporal constraints used for peak validation. Detected peaks were required to be local maxima and separated by at least 200 ms to avoid duplicate detections and reduce noise-induced false positives. Cadence was computed from the median inter-peak interval, and mean repetition time was taken directly from this median.

Active time was defined as the total duration during which the gravity-compensated and smoothed accelerometer magnitude exceeded 0.15 g (1.5 $\mathrm{m/s^2}$). This threshold was selected based on the observation that light human motion typically remains below approximately 0.2 g (47).

3.6 Synthetic Dataset Generation

To test the software of the web application without interference from IMU errors, synthetic data was created and investigated. For this, three synthetic IMU datasets were generated at 100 Hz from analytically defined ground-truth trajectories. Each trajectory was first specified as a sequence of time-stamped position and orientation segments. Synthetic accelerometer, gyroscope, and magnetometer readings were derived by differentiating the trajectory and rotating the relevant physical quantities into the sensor frame. Independent Gaussian noise was added to each sensor axis with $0.05\,\mathrm{m\,s^{-2}}$ RMS on each accelerometer axis, $0.3\,^\circ\mathrm{s^{-1}}$ RMS on each gyroscope axis, and $0.5\,\mu\mathrm{T}$ RMS on each magnetometer axis. The magnetometer reference field was configured with $B_N = 20\,\mu\mathrm{T}$ and $B_D = 45\,\mu\mathrm{T}$, wherein the horizontal component rotates with yaw. All the synthetic datasets begin and end with a 5 s rest phase to allow the gyroscope bias estimator to converge.

Dataset 1 — Pure Linear Motion. A sequence of translational bursts along each of the three axes produces a total displacement of 16 m and a peak speed of $8\,\mathrm{m\,s^{-1}}$ with no change in orientation. Ground-truth rotation metrics are zero on all three rotation axes, and the total duration is approximately 18 s.

Dataset 2 — Pure Rotational Motion. Three sequential rotation sweeps cover 180° in yaw, 60° in pitch, and 60° in roll with no translational acceleration. Ground-truth distance and speed are zero throughout, and the total duration is approximately 16 s.

Dataset 3 — Mixed Linear and Rotational Motion. Three paired phases combine simultaneous translation and rotation. Phase A pairs forward linear motion (4 m, $5\,\mathrm{m\,s^{-1}}$ peak) with a 90° yaw sweep, Phase B pairs vertical motion (5 m, $5\,\mathrm{m\,s^{-1}}$) with a 60° pitch sweep, and Phase C pairs lateral motion (4 m, $5\,\mathrm{m\,s^{-1}}$) with a 60° roll sweep. The total ground-truth displacement is 13 m and peak speed is $5\,\mathrm{m\,s^{-1}}$, and the total duration is approximately 18 s.

3.7 Evaluation Metrics

The performance of each pipeline was characterised by six scalar metrics extracted from the output for comparison against the known ground truth. *Total distance* is the final value of the cumulative speed integral. *Peak speed* is the maximum instantaneous speed across the recording. *Yaw, pitch, and roll excursion* are the total signed angular displacement on each axis between the first and last sample. *Peak linear acceleration residual* (Dataset 2 only) is the maximum magnitude of the gravity-removed accelerometer signal during active rotation phases. Errors are reported as signed deviations and as percentage errors relative to the ground-truth value.

3.8 Experimental Protocol

Three distinct experimental stages were conducted in this investigation. First, the three pipelines were compared with the synthetic data against each other and the ground truth to identify an ideal candidate pipeline. Second, this candidate pipeline underwent a sensitivity study to investigate the optimal parameter configuration for accuracy. Third, this final ideal pipeline was validated against real IMU data to investigate performance in a real world scenario and identify magnitudes of IMU error.

3.8.1 Sensor Fusion Algorithm Comparison

All three pipelines were executed on each synthetic dataset with the following fixed parameters. The Expanded Madgwick gain was $\beta = 0.033$, the standard default providing a balance between noise rejection and convergence speed appropriate for moderate dynamic motion. The EKF measurement gate threshold was $\pm 0.2\,g$, chosen to admit accelerometer updates under near-static conditions while rejecting samples where translational acceleration is likely to dominate. The VQF parameters were $\tau_{\text{acc}} = 5\,\text{s}$ and $\tau_{\text{mag}} = 9\,\text{s}$, selected as an intermediate starting point ahead of the sensitivity study, providing moderate rejection of translational acceleration and responsive heading correction. Gyroscope bias calibration and ZUPT detection were enabled for all pipelines with the thresholds as stated above in Section 3.4. A 50-sample warm-up period was excluded from all metric computations.

3.8.2 VQF Parameter Sensitivity Study

Following algorithm selection, a full-factorial grid search over $\tau_{\text{acc}} \in \{1, 2, 3, 5, 10, 15, 20\}\,\text{s}$ and $\tau_{\text{mag}} \in \{3, 6, 9, 15, 30, 60, 90\}\,\text{s}$ was conducted on all three synthetic datasets, yielding $7 \times 7 = 49$ combinations per dataset and $49 \times 3 = 147$ runs in total. All other pipeline settings were held fixed at the values used in the algorithm comparison.

3.8.3 VQF Pipeline Validation Experiment

Two physical experiments were conducted to test the VQF pipeline against known, real-world ground truths: a pure rotation experiment and a mixed translational-rotational experiment. A linear motion experiment was also attempted but was not pursued due to the fundamental limitations of double integration at low accelerations. This is discussed in Section 5.5.1. All experiments used the optimised VQF configuration identified in the parameter sensitivity study ($\tau_{\text{acc}} = 15\,\text{s}$, $\tau_{\text{mag}} = 60\,\text{s}$). The sensor used throughout all real experiments was a Movesense model OP174 IMU connected to the Movesense Showcase application on a smartphone (seen in Appendix A Figure A1), configured with a $\pm 4\,g$ accelerometer range and $\pm 1000\,^{\circ}\text{s}^{-1}$ gyroscope range at 104 Hz. For all real recordings, a low-pass filter cutoff of 10 Hz was applied to accelerometer and gyroscope signals prior to fusion, with a secondary 5 Hz cutoff applied before velocity integration. These cutoffs were selected in accordance with the 10 Hz upper bound

of typical human motion content established in Section 2.4.1, providing more aggressive noise suppression than the 20 Hz cutoff applied to the synthetic datasets.

To facilitate analysis of specific movement phases within each recording, a range analysis tool was developed as part of the MoveSync web application. This tool allows a time window to be selected over a recording, with the same motion metrics computed over that range as those used in the synthetic studies. An additional option to suppress automatic ZUPT detection within the selected range was also implemented. Where the motion within a range is known, disabling ZUPT detection prevents false triggers that would otherwise reset velocity during genuine movement, allowing the pipeline to integrate freely across the full range. This suppression was enabled for the mixed experiment but not the rotation experiment, where genuine stationary holds between steps meant ZUPT detection behaved correctly.

Rotation Experiment Setup The rotation experiment was designed to assess the pipeline's orientation tracking against a known ground truth of 90° per step with zero translational displacement. The IMU was placed flat face-down on a smooth hard surface. Marks were made at 0°, 90°, and 180° around the device and an identifiable feature on the IMU rim was aligned with the 0° reference line. This setup can be seen in Appendix A Figure A2. A fingertip placed centrally on the IMU from above was used as an informal pivot constraint during rotation. This does not provide a rigid constraint and so some small translational displacement may be present, discussed in Section 5.5.2. Each trial comprised two sequential 90° yaw steps: the IMU was rotated from 0° to 90°, held stationary for approximately two seconds, then rotated from 90° to 180°, followed by a final stationary hold. Recordings were kept short to minimise any heading drift from environmental magnetometer disturbance. Three trials were conducted, yielding six individual 90° rotation measurements.

Mixed Motion Experiment Setup To recreate a mixed motion dataset combining translational and rotational movement, an empty glass jar was used as a rolling platform, chosen over metal to avoid magnetometer interference. The ground truth distance of 0.71 m for three full rotations was measured directly on the rolling surface. The IMU was mounted centrally on the flat end face of the jar and secured with tape so that it could not slip during rolling. This setup can be seen in Appendix A Figure A2. This placement ensured the IMU underwent no tumbling motion as the jar rolled, remaining at a fixed height with its motion confined to horizontal translation and yaw rotation. A reference line was marked on the jar with a corresponding start and finish line marked on the rolling surface to allow full rotations to be counted visually.

Three trials were conducted. For each trial the jar was positioned at the start line with the reference marks aligned. A stationary hold of at least five seconds was recorded, after which the jar was rolled steadily forward for three full rotations. The jar was then stopped and held stationary for a further five seconds before the recording ended. Ground truth yaw rotation was $3 \times 360° = 1080°$, assuming perfect rolling contact without slip. Any slip during rolling would introduce a small error into these ground truth values.

4 Results

4.1 Web Application Progress

The MoveSync web application was implemented and published through a GitHub repository (48). The repository contains the complete source code and provides access to the deployed version of the application (49). This allows the implementation to be inspected, the application setup to be reproduced, and the final web-based system to be tested through the deployment link.

The final implementation consists of an interconnected set of modules for project management, data import, session storage, signal visualization, data processing, and session comparison. The overall application workflow is presented in Figure 6, illustrating the relationship between the main modules and the sequence of actions available to the user.

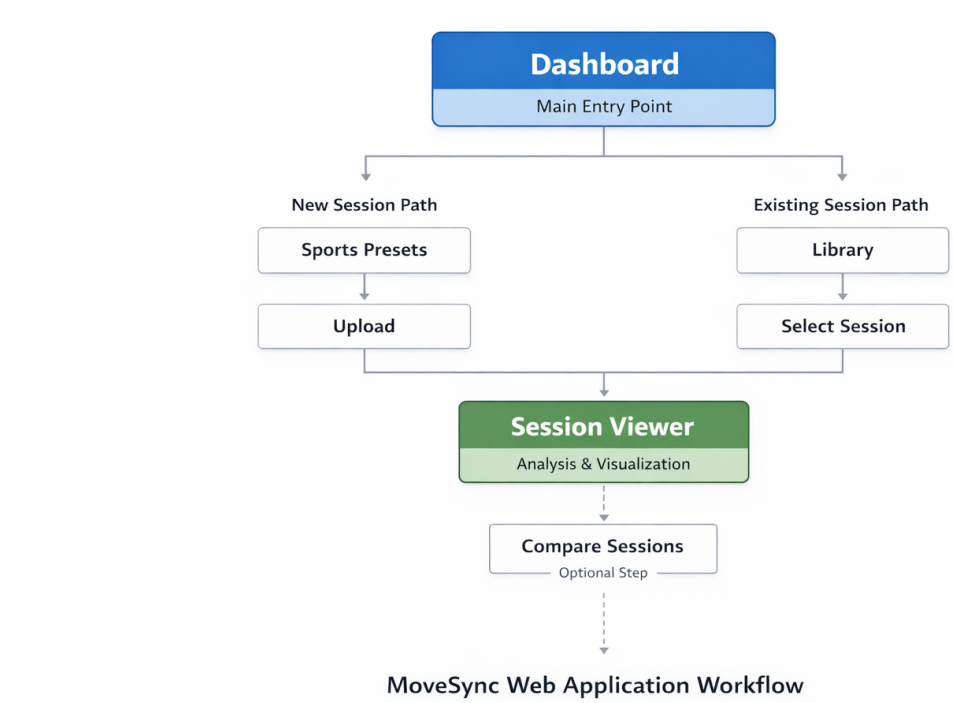


Figure 6: Conceptual overview of the MoveSync web application workflow, showing the connection between project management, data upload, session inspection, processing, and comparison.

The *Dashboard* module provides the initial overview of the implemented system. As shown in Figure 7, the dashboard displays the number of projects and sessions, recent activity, and the current local storage status. This module, therefore, functions as the main entry point for monitoring the stored data and navigating to the main parts of the application.

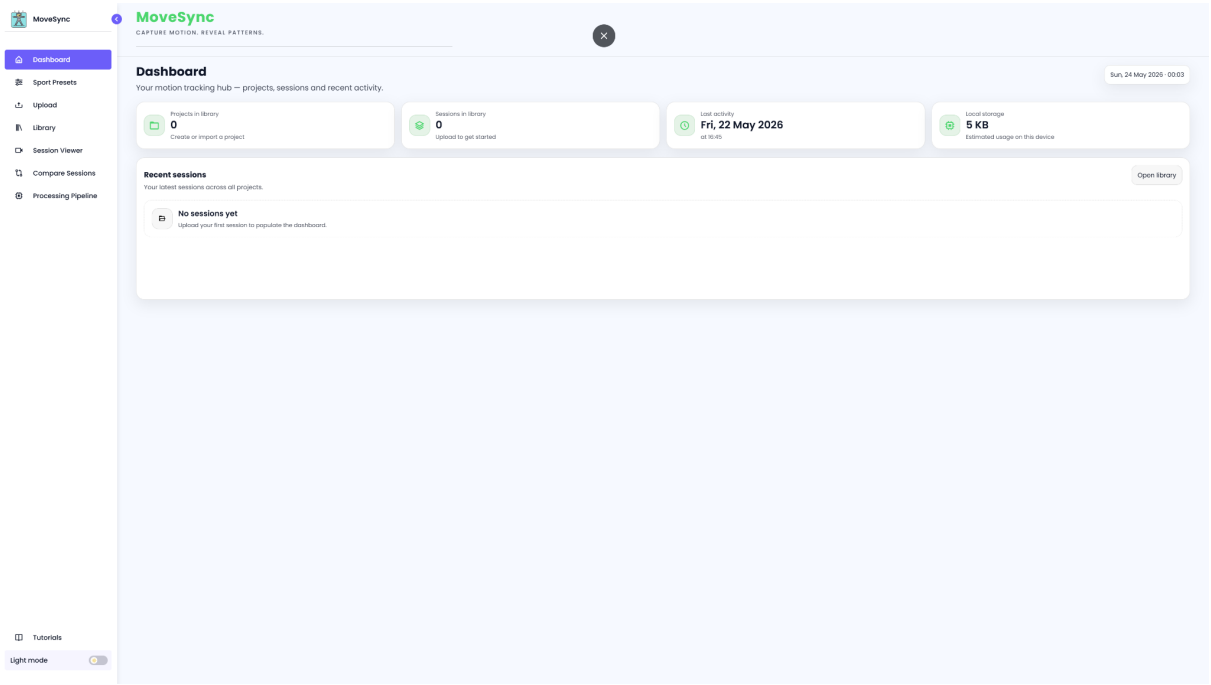


Figure 7: Dashboard module showing an overview of projects, recorded sessions, recent activity, and local storage status in the MoveSync web application.

The *Sport Presets* module contains the implemented preset configurations for different motion-analysis tasks. Figure 8 shows how activity-specific presets are presented together with selected metrics, timestamp categories, and analysis windows. These presets define the analysis settings that can be used when evaluating motion-tracking sessions.

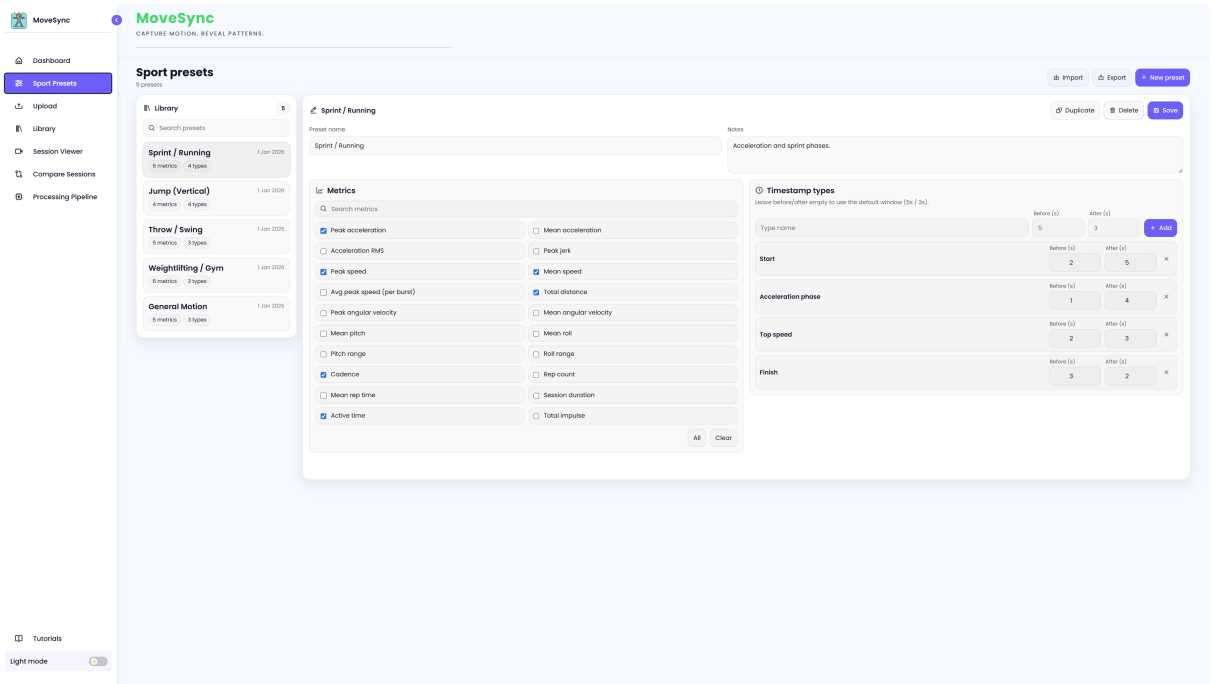


Figure 8: Sport Presets module used to define activity-specific analysis configurations, including selected metrics, timestamp categories, and analysis windows.

The *Upload* module provides the implemented interface for importing new motion-tracking data or previously saved projects into the system. As presented in Figure 9, the module includes fields for project creation and file selection. The interface supports the upload of both video recordings and corresponding IMU data files, which are required for synchronized motion analysis.

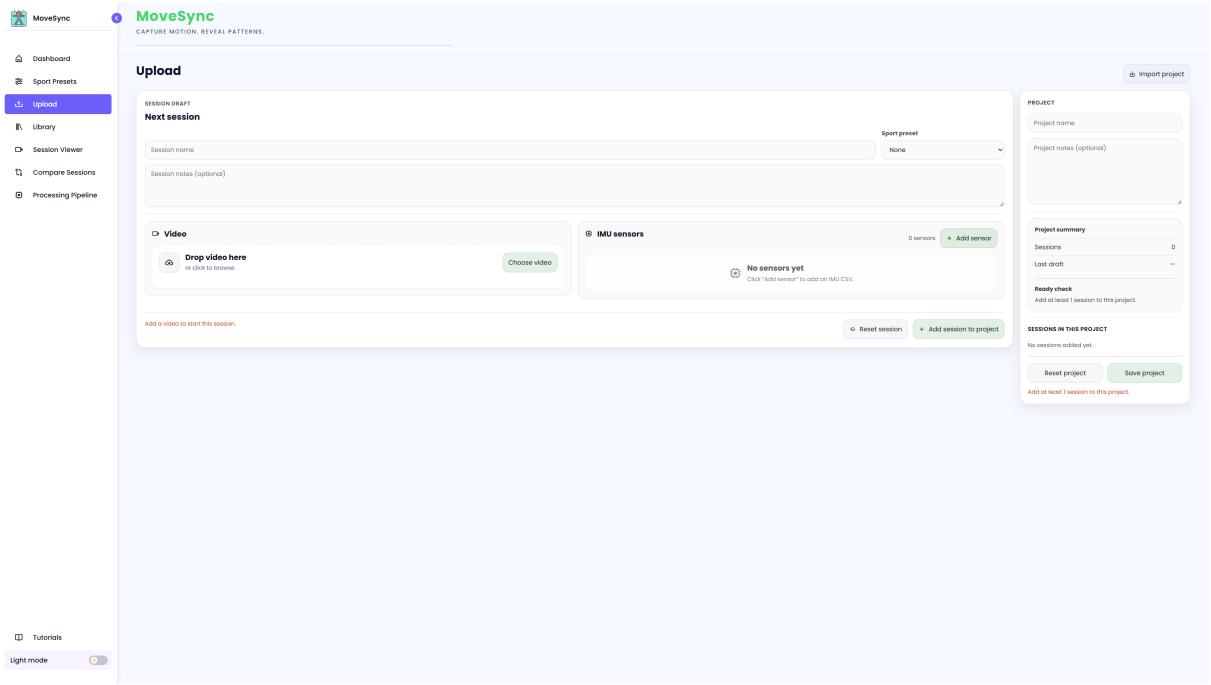


Figure 9: Upload module for creating a new project and importing video recordings together with corresponding IMU data files.

The *Library* module presents the projects (containing sessions) stored in the application. Figure 10 shows the resulting data structure, including session information and available actions for opening, editing, exporting, or deleting stored recordings. This module provides a centralized view of the available data and supports continued analysis of previously imported projects.

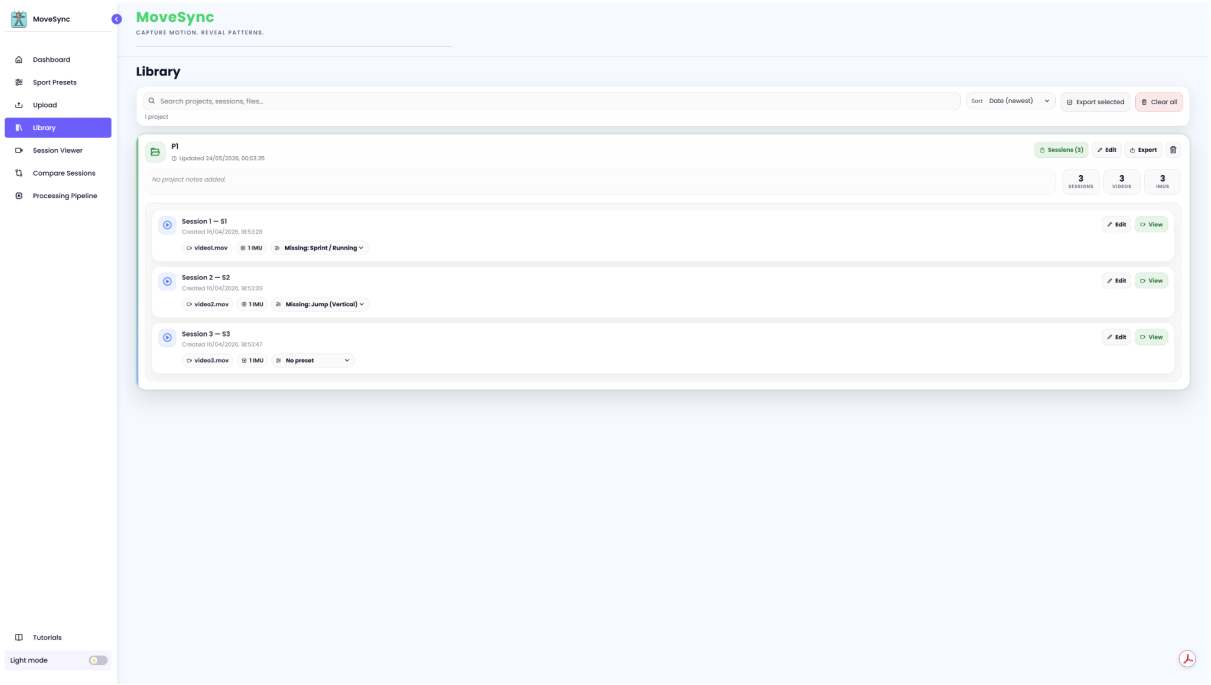


Figure 10: Library module providing a centralized overview of stored projects and sessions, with options for retrieval, editing, export, and further analysis.

The *Session Viewer* module is the main interface for inspecting an individual motion-tracking session. As shown in Figure 11, the implemented view includes synchronized video playback, IMU-derived signal plots, session information, and a metrics display for the selected recording. The lower part of the interface contains separate tabs for IMU graphs, expanded analysis with extracted metrics, and Sensor Fusion 3D visualization.

The module also includes MoveNet-based motion tracking, which enables real-time detection of body landmarks during video playback. This functionality is used in the angle analysis tool, where the angle between three selected body points is displayed dynamically throughout the movement. In addition, the Sensor Fusion 3D tab presents a three-dimensional visualization of IMU-based orientation data, providing an additional representation of the recorded motion alongside the video and signal-based results.

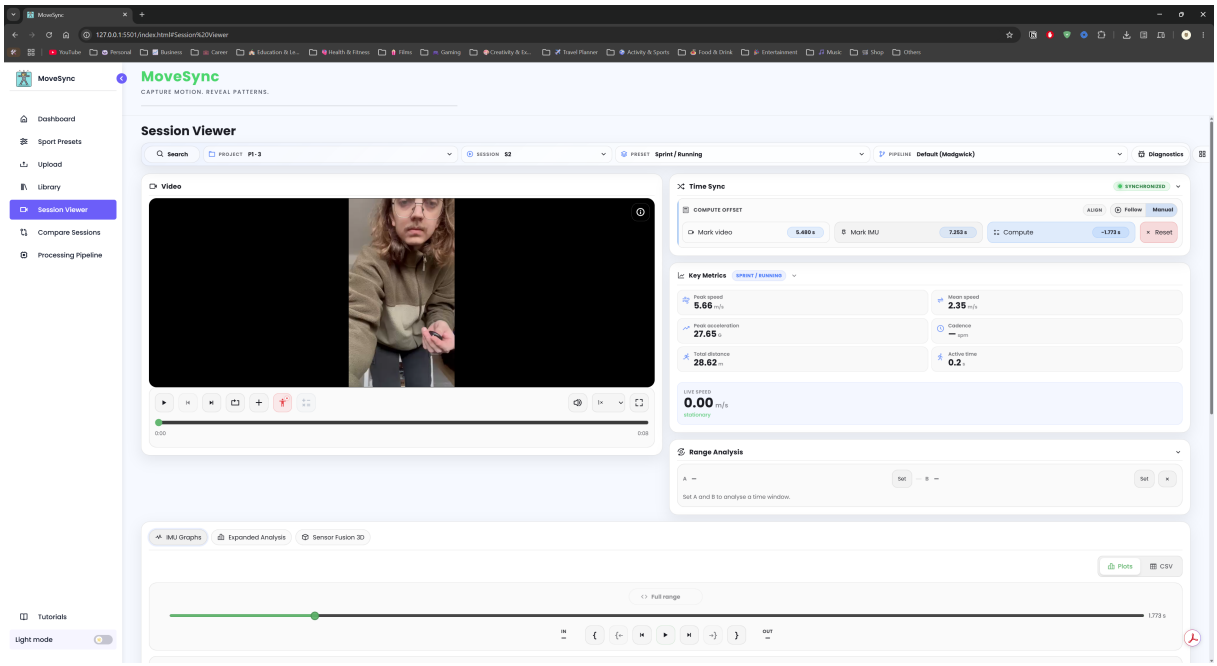


Figure 11: Session Viewer module showing synchronized video playback, IMU-derived signal plots, MoveNet-based angle analysis, extracted session metrics, expanded analysis, and Sensor Fusion 3D visualization for inspection of an individual recording.

The *Compare Sessions* module provides the implemented interface for comparing two motion-tracking recordings. The video comparison view is presented in Figure 12. In this view, two selected sessions are displayed side by side with synchronized video windows and corresponding session information, enabling direct visual comparison of the recorded movements.

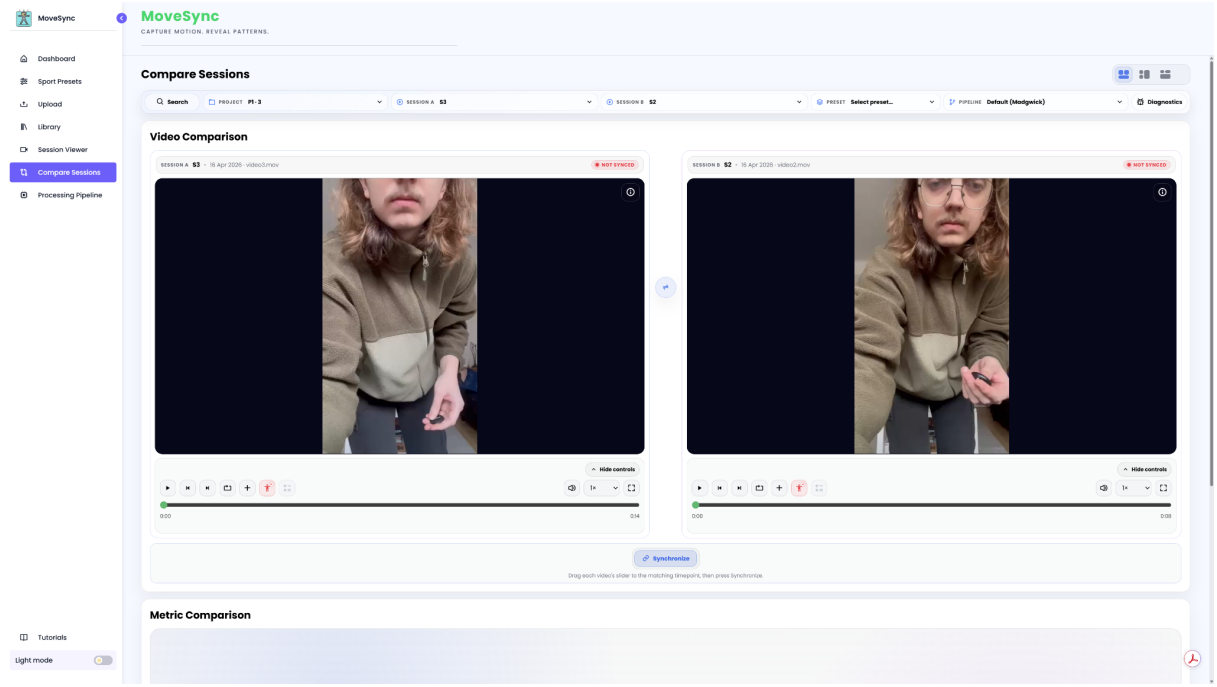


Figure 12: Compare Sessions module showing the video comparison view, where two motion-tracking sessions are displayed side by side for visual inspection.

The module also includes a metric comparison view for evaluating extracted performance

measures between sessions. As shown in Figure 13, the interface presents comparison results using metric summaries and visualized differences between the selected recordings. This allows quantitative differences between sessions to be reviewed together with the visual comparison.

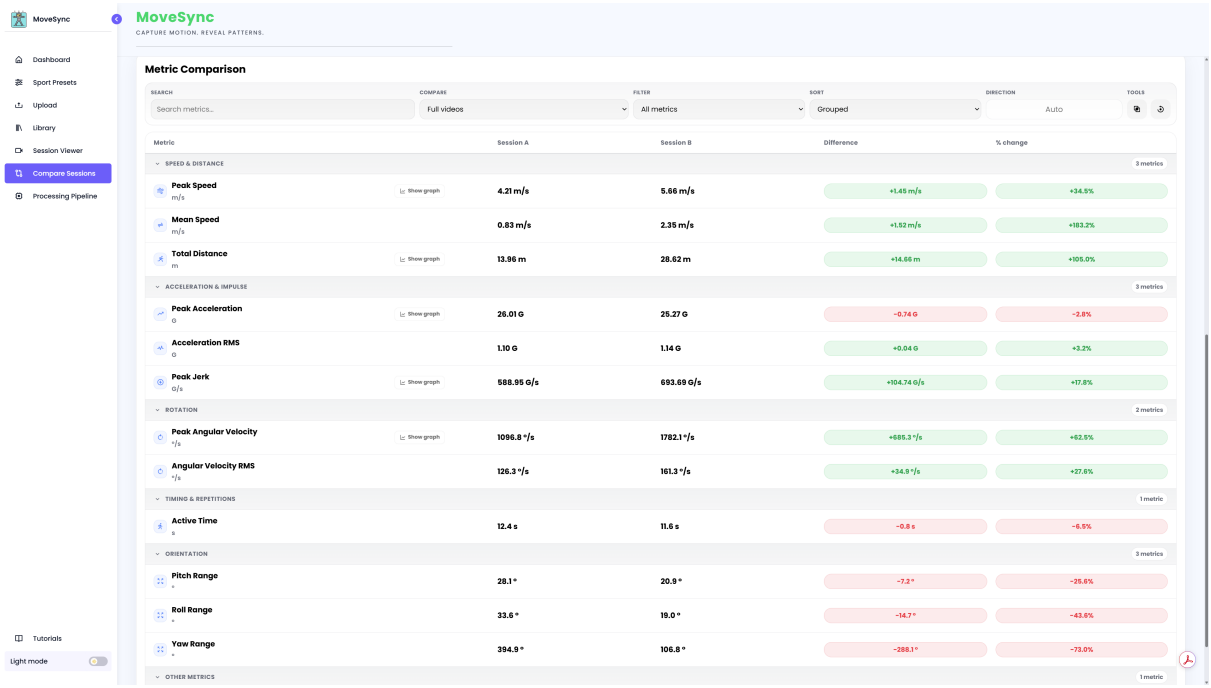


Figure 13: Compare Sessions module showing the metric comparison view, where extracted performance measures from two sessions are presented for comparison.

The *Processing Pipeline* module contains the implemented controls for preprocessing and sensor-fusion configuration. As shown in Figure 14, the interface includes settings for filtering, calibration, zero-velocity updates, and processing parameters. These controls define how the imported IMU data are prepared before visualization and analysis.

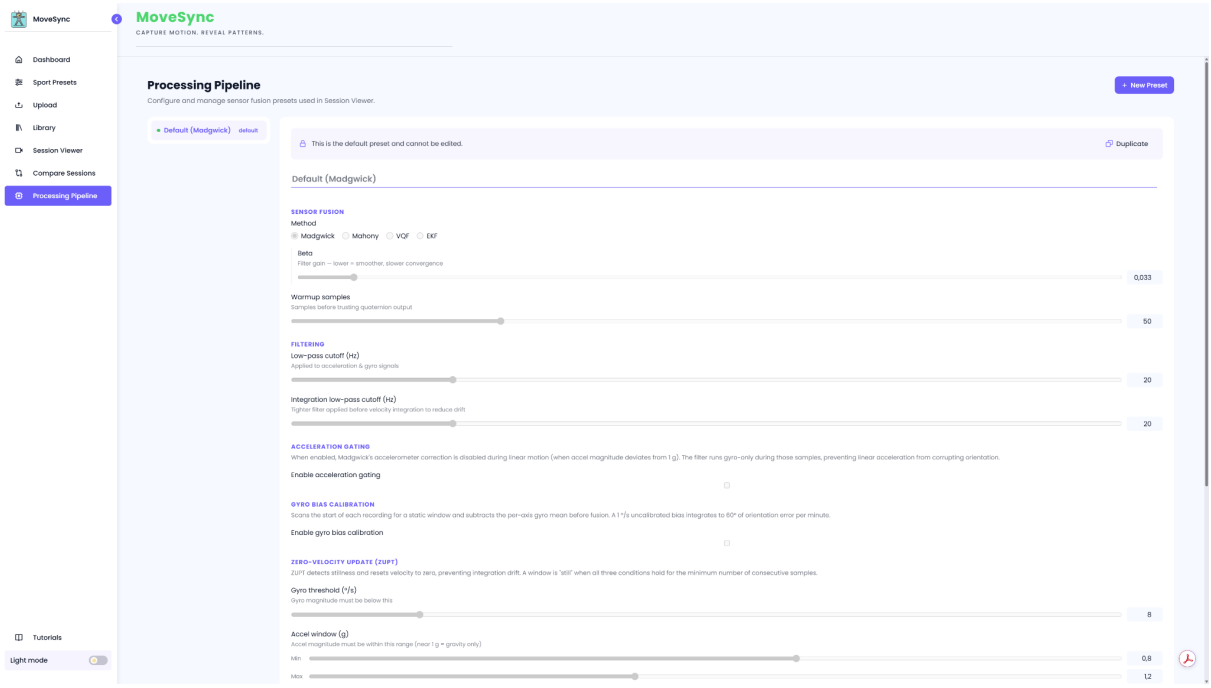


Figure 14: Processing Pipeline module containing configurable preprocessing and sensor-fusion settings used to prepare IMU data for analysis.

The *Tutorials* interface provides the instructional component of the final web application. Figure 15 shows the integrated guidance material, including sections for project setup, data upload, session viewing, comparison, and advanced analysis. This interface supports the use of the application by explaining the main workflow and available functions.

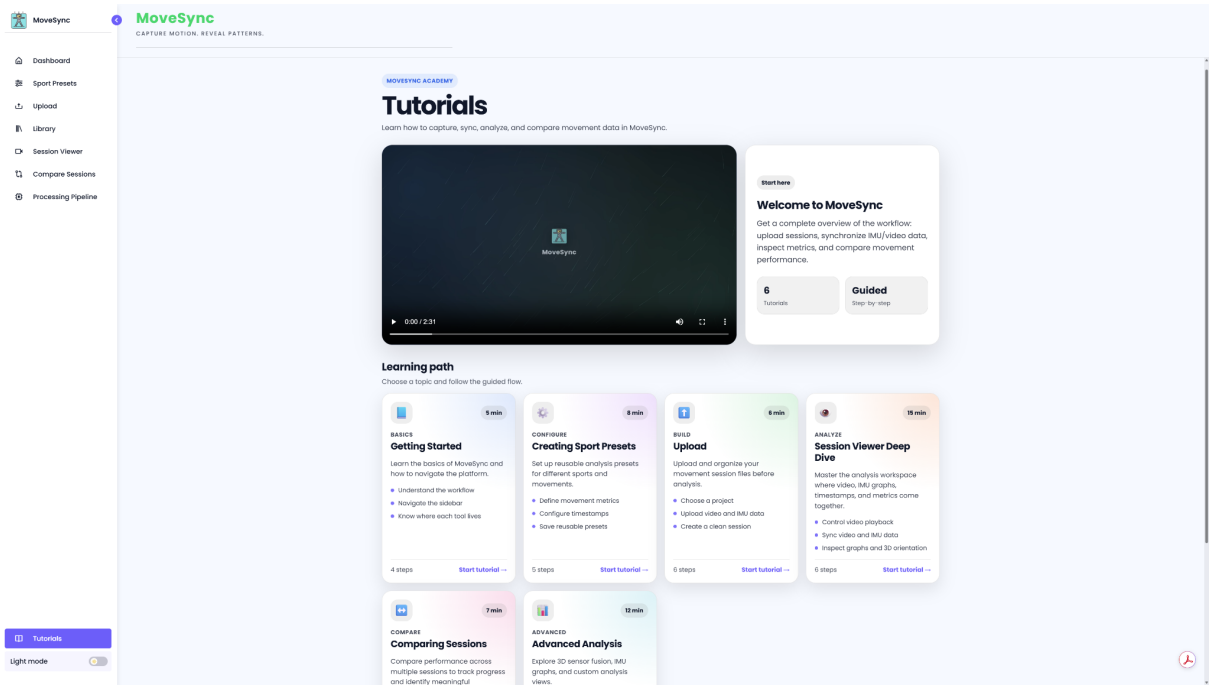


Figure 15: Tutorials interface providing instructional guidance for using the MoveSync web application and understanding the main workflow.

4.2 Algorithm Comparison on Synthetic Datasets

Three sensor fusion pipelines were evaluated on the synthetic datasets described in Section 3.6: Dataset 1 (pure linear motion), Dataset 2 (pure rotation), and Dataset 3 (mixed motion). Each pipeline was assessed against known ground truth values for distance, peak speed, and orientation excursion on all three axes.

4.2.1 Dataset 1 — Pure Linear Motion

The pipeline outputs are summarised in Table 1. Expanded Madgwick produced the closest distance estimate (15.57 m, −3 %) and competitive peak speed recovery (7.87 $\mathrm{m\,s^{-1}}$, −2 %), but accumulated the largest orientation errors of the three pipelines: yaw 46°, pitch 6°, and roll 2° against ground truth values of 0°.

EKF returned a distance estimate of 15.06 m (−6 %) with equivalent speed recovery (7.95 $\mathrm{m\,s^{-1}}$, −1 %) and tight yaw and pitch tracking (3° each), though it had the largest roll error of the three pipelines (9°). VQF achieved the smallest yaw error (2°) and intermediate pitch and roll errors (8° and 6° respectively), but underestimated distance (14.31 m, −11 %) and peak speed (7.34 $\mathrm{m\,s^{-1}}$, −8 %).

Table 1: Pipeline outputs on Dataset 1 (pure linear) vs. ground truth (GT: distance = 16.0 m, peak speed = 8.0 $\mathrm{m\,s^{-1}}$, all rotation metrics = 0°).

Pipeline	Dist. (m)	Dist. err.	Peak v ($\mathrm{m\,s^{-1}}$)	Speed err.	Yaw (°)	Pitch (°)	Roll (°)
Ground truth	16.0	—	8.00	—	0	0	0
Exp. Madgwick	15.57	−3%	7.87	−2%	46	6	2
VQF ($\tau_{\mathrm{acc}} = 5\,\mathrm{s}$)	14.31	−11%	7.34	−8%	2	8	6
EKF	15.06	−6%	7.95	−1%	3	3	9

4.2.2 Dataset 2 — Pure Rotation

The summary metrics for Dataset 2 are given in Table 2. All three pipelines correctly output zero speed and zero distance throughout the recording, confirming that the rotation gate successfully prevented phantom velocity accumulation during high-rate gyroscope motion.

The orientation accuracy diverged sharply between the pipelines. Expanded Madgwick accumulated a yaw range of 1293° against a ground truth of 180°, with pitch and roll ranges of 129° and 209° against ground truth values of 60°.

EKF and VQF both tracked all three rotation axes accurately. EKF returned yaw 180.4° (+0.4° error), pitch 60.1° (+0.1°), and roll 60.3° (+0.3°). VQF estimated yaw 189.2° (+9.2°), pitch 60.1°, and roll 60.2°. The peak linear acceleration residual during active rotation was 19.75 $\mathrm{m\,s^{-2}}$ for Expanded Madgwick against 0.22 $\mathrm{m\,s^{-2}}$ for EKF and 0.23 $\mathrm{m\,s^{-2}}$ for VQF, a ≈90× difference.

Table 2: Pipeline outputs on Dataset 2 (pure rotation) vs. ground truth (GT: yaw = 180°, pitch = 60°, roll = 60°). Speed and distance were zero for all pipelines.

Pipeline	Yaw (°)	Yaw err.	Pitch (°)	Pitch err.	Roll (°)	Roll err.	Peak a_{lin} ($\mathrm{m\,s^{-2}}$)
Ground truth	180	—	60.0	—	60.0	—	≈ 0
Exp. Madgwick	1293	+618%	129	+115%	209	+248%	19.75
VQF ($\tau_{\mathrm{acc}} = 5\,\mathrm{s}$)	189.2	+5%	60.1	< 1%	60.2	< 1%	0.23
EKF	180.4	< 1%	60.1	< 1%	60.3	< 1%	0.22

4.2.3 Dataset 3 — Mixed Motion

The summary metrics for Dataset 3 are given in Table 3. The dataset comprises three phases: forward motion with yaw (Phase A), vertical motion with pitch (Phase B), and lateral motion with roll (Phase C).

Table 3: Pipeline outputs on Dataset 3 (mixed motion) vs. ground truth (GT: distance = 13.0 m, peak speed = 5.0 m s^{-1}, yaw = 90°, pitch = 60°, roll = 60°). Percentage errors shown in parentheses.

Pipeline	**Dist. (m)**	**Peak v (m s^{-1})**	**Yaw (°)**	**Pitch (°)**	**Roll (°)**
Ground truth	13.0	5.00	90.0	60.0	60.0
Exp. Madgwick	13.98 (+8%)	5.83 (+17%)	430 (+378%)	90.4 (+51%)	147 (+145%)
VQF ($\tau_{\text{acc}} = 5$ s)	11.60 (−11%)	4.51 (−10%)	93.6 (+4%)	60.6 (+1%)	60.9 (+1%)
EKF	11.21 (−14%)	4.02 (−20%)	86.9 (−3%)	60.7 (+1%)	67.5 (+13%)

Expanded Madgwick overestimated the total distance by 8 % (13.98 m vs. 13.0 m ground truth) and peak speed by 17 % (5.83 m s^{-1} vs. 5.0 m s^{-1}), with orientation errors of 430° in yaw, 90° in pitch, and 147° in roll against ground truth values of 90°, 60°, and 60° respectively.

VQF produced the most accurate results overall: distance 11.60 m (−11 %), peak speed 4.51 m s^{-1} (−10 %), yaw 93.6° (+4 %), pitch 60.6° (+1 %), and roll 60.9° (+1 %). EKF underestimated distance more severely (11.21 m, −14 %) and peak speed by 20 % (4.02 m s^{-1}), with a roll error of +13 % (67.5°).

4.2.4 Algorithm Selection for Parameter Study

Across all three datasets, VQF and EKF both demonstrated accurate and consistent performance, clearly outperforming Expanded Madgwick on any scenario involving rotation. For Dataset 1, EKF achieved the most accurate distance estimate (−6 %) and the most accurate yaw tracking on Dataset 2 (+0.4° vs. +9.2° for VQF). However, on Dataset 3 the gap between the two algorithms widened considerably: VQF returned a peak speed error of −10 % against EKF's −20 %, and a roll error of +1 % against EKF's +13 %. Expanded Madgwick performed unacceptably across all rotation scenarios.

VQF is therefore selected as the ideal pipeline and is the basis for the parameter sensitivity study in Section 3.8.2, in which τ_{acc} and τ_{mag} are varied to characterise their effect on the speed–orientation accuracy trade-off.

4.3 VQF Parameter Sensitivity Study

The results of the grid search described in Section 3.8.2 are presented below. Results are organised by the effect of each parameter independently, before a comparison of the default and recommended configurations is given.

4.3.1 Parameter Orthogonality

The two time constants are structurally orthogonal: across all parameter combinations, varying τ_{mag} produced no measurable change in distance, speed, pitch, or roll, and varying τ_{acc} produced no measurable change in yaw. The two parameters can therefore be tuned independently without cross-effects.

4.3.2 Effect of $\tau_{\mathbf{acc}}$

Table 4 shows the effect of τ_{acc} on kinematic and tilt accuracy across the linear and mixed datasets, with τ_{mag} fixed at 9 s. On the pure rotation dataset, τ_{acc} had no effect on any metric. All metrics improve uniformly with increasing τ_{acc} across both datasets, though with clear diminishing returns. For example, the improvement in mixed distance from $\tau_{\text{acc}} = 10$ s to 15 s is 0.21 m, compared to 0.11 m from 15 s to 20 s.

Table 4: Effect of τ_{acc} on kinematic and tilt accuracy. $\tau_{\text{mag}} = 9$ s throughout. GT: linear 16.0 m, 8.0 $\mathrm{m\,s^{-1}}$; mixed 13.0 m, 5.0 $\mathrm{m\,s^{-1}}$. Pitch and roll ground truth is 0° for the linear dataset.

τ_{acc} (s)	Linear		Mixed		Mixed orientation	
	Dist (m)	Speed ($\mathrm{m\,s^{-1}}$)	Dist (m)	Speed ($\mathrm{m\,s^{-1}}$)	Pitch (°)	Roll (°)
1	11.59	5.46	9.75	3.97	68.80	70.70
2	12.82	6.52	10.63	3.97	63.35	63.93
3	13.54	6.96	11.10	4.25	61.69	61.85
5	14.31	7.34	11.60	4.51	60.61	60.85
10	15.04	7.66	12.11	4.72	60.32	60.23
15	15.33	7.76	12.32	4.79	60.32	60.17
20	15.49	7.82	12.43	4.83	60.33	60.21

4.3.3 Effect of $\tau_{\mathbf{mag}}$

Table 5 shows the effect of τ_{mag} on yaw accuracy. Since τ_{acc} does not influence yaw, results from the pure rotation dataset are independent of τ_{acc} and a single column suffices. Yaw error decreases rapidly with increasing τ_{mag} up to 30 s before flattening sharply. The improvement from 30 s to 60 s is 1.5°, reducing further to just 0.5° from 60 s to 90 s.

Table 5: Effect of τ_{mag} on yaw accuracy. GT: rotation 180°, mixed 90°. τ_{acc} does not affect yaw.

τ_{mag} (s)	Rotation yaw (°)	Rotation error (°)	Mixed yaw (°)	Mixed error (°)
3	205.83	+25.8	100.15	+10.2
6	193.67	+13.7	95.25	+5.3
9	189.17	+9.2	93.58	+3.6
15	185.39	+5.4	92.23	+2.2
30	182.45	+2.5	91.22	+1.2
60	180.95	+1.0	90.71	+0.7
90	180.44	+0.4	90.54	+0.5

4.3.4 Optimal Configuration

Table 6 compares the default configuration used in the algorithm comparison ($\tau_{\text{acc}} = 5$ s, $\tau_{\text{mag}} = 9$ s) against a practically optimal configuration ($\tau_{\text{acc}} = 15$ s, $\tau_{\text{mag}} = 60$ s) selected on diminishing-returns grounds. The rationale for not extending to the search boundaries is discussed in Section 5.3.2.

Table 6: Default vs practically optimal VQF configuration across all three datasets. Error reduction is the absolute reduction in error magnitude.

Metric	Unit	GT	Default $(5, 9)$	Recommended $(15, 60)$	Error reduction
Linear distance	m	16.00	14.31 (−11%)	15.33 (−4%)	1.02
Linear speed	$\mathrm{m\,s^{-1}}$	8.00	7.34 (−8%)	7.76 (−3%)	0.42
Mixed distance	m	13.00	11.60 (−11%)	12.32 (−5%)	0.72
Mixed speed	$\mathrm{m\,s^{-1}}$	5.00	4.51 (−10%)	4.79 (−4%)	0.28
Mixed yaw	°	90.0	93.58 (+3.6°)	90.71 (+0.7°)	2.9
Mixed pitch	°	60.0	60.61	60.32	0.29
Mixed roll	°	60.0	60.85	60.17	0.68
Rotation yaw	°	180.0	189.17 (+9.2°)	180.95 (+1.0°)	8.2
Rotation pitch	°	60.0	60.07	60.07	—
Rotation roll	°	60.0	60.17	60.17	—

Every metric improves or remains unchanged under the recommended configuration. The largest gains are in rotation yaw (8.2° error reduction) and linear distance (1.02 m). Pitch and roll on the rotation dataset are unaffected by either time constant.

4.4 Real Experimental Results

Results from both real-world validation experiments are presented in this section. Rotation performance is first evaluated using the controlled 90° step experiment, followed by analysis of the mixed translational-rotational rolling experiment.

4.4.1 Rotation Results

Results for all six 90° rotation steps across three trials are given in Table 7. All steps produced near-zero distance estimates, with a maximum of 0.01 m, confirming that automatic ZUPT detection successfully suppressed phantom velocity throughout all rotation phases.

Yaw tracking was accurate and highly consistent. All six measurements overshot the 90° ground truth by a small systematic margin, with individual errors ranging from +3.4% to +7.5% and a mean yaw of 94.67° across all measurements (+5.2% mean error). The overshoot was consistent across trials, indicating a systematic rather than stochastic bias. Pitch range remained small throughout, with a mean of 3.92° across all six measurements, consistent with minor physical tilt during the manual pivot. Roll range was systematically higher during the second spin of each trial (10.80° mean) than the first (5.75° mean), reproducible across all three trials. This asymmetry is discussed in Section 5.5.2.

4.4.2 Mixed Motion Results

Results for the three trials are given in Table 8. Yaw tracking was highly accurate, with a mean of 1088.90° against the 1080° ground truth (+0.82% mean error), comparable to the per-step yaw accuracy observed in the rotation experiment (+5.2% per 90° step). Distance estimation was substantially larger than ground truth in all trials, with a mean estimated distance of 4.18 m against 0.71 m (+489% error). This result is consistent with the expected behaviour of double integration under low-acceleration conditions and is discussed further in Section 5.5.3.

Table 7: Pipeline outputs for six individual 90° yaw steps across three trials (GT: yaw = 90°, pitch = 0°, roll = 0°, distance = 0 m).

Trial	Step	Yaw (°)	Yaw err.	Pitch (°)	Roll (°)	Dist. (m)	Peak gyro (° s^{-1})
Ground truth	—	90.0	—	0	0	0.00	—
1	Spin 1	94.56	+5.1%	3.90	4.30	0.01	170.08
1	Spin 2	95.29	+5.9%	2.98	9.58	0.00	203.35
2	Spin 1	94.86	+5.4%	4.18	4.89	0.00	118.32
2	Spin 2	93.55	+3.9%	3.59	11.84	0.00	150.35
3	Spin 1	93.02	+3.4%	4.39	8.06	0.00	133.60
3	Spin 2	96.76	+7.5%	4.45	10.98	0.00	189.43
Mean		**94.67**	**+5.2%**	**3.92**	**8.28**	**0.00**	**160.86**

Table 8: Pipeline outputs for three jar-rolling trials (GT: yaw = 1080°, distance = 0.71 m).

Trial	Yaw (°)	Yaw err.	Distance (m)	Peak gyro (° s^{-1})
Ground truth	1080.0	—	0.71	—
1	1098.93	+1.75%	4.52	518.00
2	1086.27	+0.58%	4.27	578.65
3	1081.51	+0.14%	3.76	495.76
Mean	**1088.90**	**+0.82%**	**4.18**	**530.80**

5 Discussion

5.1 System Workflow and Usability

The implemented MoveSync web application demonstrates that the main system workflow has reached a functional prototype stage. The application integrates project creation, data upload, session storage, preprocessing, visualization, and session comparison within a single web-based interface. This integrated structure improves workflow continuity, since users can move between the main stages of the analysis without relying on separate tools for file organization, signal inspection, processing, or comparison.

5.1.1 Workflow Structure and Navigation

The modular design supports usability by separating the application into clearly defined functional parts. The *Dashboard* and *Library* modules provide an overview of stored projects and sessions, while the *Upload* and *Sport Presets* modules support the preparation of new analysis sessions. This division makes the workflow easier to follow and reduces the amount of repeated manual setup required before analysis.

The *Tutorials* interface further strengthens usability by providing integrated guidance within the application. This is particularly important for users who are unfamiliar with the system workflow, since the application includes several stages. By placing the guidance material directly in the web application, users can access instructions without leaving the system environment.

5.1.2 Session Inspection and Analysis

The *Session Viewer* represents the most developed analysis component of the web application. By combining synchronized video playback, IMU-derived signal plots, extracted session metrics, MoveNet-based angle analysis, and Sensor Fusion 3D visualization, the module provides several complementary views of the same recording. This is important because movement analysis often benefits from both qualitative visual inspection and quantitative signal-based information.

The use of separate tabs for IMU graphs, expanded analysis, and Sensor Fusion 3D visualization also improves the structure of the interface. Instead of presenting all information in one crowded view, the results are divided into categories that can be inspected individually. This makes the interface more organized and allows users to focus on specific aspects of the recording depending on the analysis task.

5.1.3 Session Comparison

The *Compare Sessions* module extends the workflow from single-session inspection to comparison between two recordings. The separation between video comparison and metric comparison is useful because it allows visual differences and quantitative differences to be evaluated in distinct views. This structure supports a more systematic comparison process, where the user can first inspect movement execution visually and then examine extracted performance measures.

However, the interpretation of session comparisons depends on the quality and consistency of the underlying data. Differences between recordings may not only reflect differences in movement performance, but may also be influenced by synchronization accuracy, sensor placement, recording conditions, preprocessing settings, and variations in how the movement was performed. Therefore, comparison results should be interpreted as dependent on both the recorded motion and the data acquisition conditions.

5.1.4 Processing Pipeline and Parameter Selection

The *Processing Pipeline* module increases transparency by exposing preprocessing and sensor-fusion parameters, including filtering, calibration, zero-velocity update settings, and other processing options. This makes the analysis workflow more flexible, since users can adjust how the IMU data are prepared before visualization and metric extraction.

At the same time, this flexibility introduces a need for careful parameter selection. Incorrect or inconsistent processing settings may affect the resulting IMU signals, extracted metrics, and Sensor Fusion 3D visualization. As a result, further testing is required to evaluate how different processing configurations influence the reliability, repeatability, and interpretability of the displayed results.

5.1.5 Synchronization and Time Offset Assumption

The synchronization approach used in the current implementation is based on the assumption that the dominant discrepancy between the video and IMU data streams is a fixed start-time offset. This assumption is reasonable when both data streams have stable sampling behavior and the main difference is caused by the recordings starting at different times.

However, this approach has limitations. If the discrepancy between the data streams changes over time, for example due to clock drift, dropped frames, inconsistent sampling intervals, or delays introduced during recording, a single time offset may not be sufficient to achieve accurate alignment throughout the full session. In such cases, additional synchronization reference points or a more advanced alignment model would be required. Future development should therefore include systematic evaluation of synchronization accuracy, especially for longer recordings and recordings collected under varying hardware or software conditions.

5.1.6 Client-Side Execution Environment

All processing in the current implementation is performed locally in a standard web browser. This includes video handling, IMU CSV parsing, synchronization, preprocessing, visualization, and session comparison. The client-side design removes the need for specialized software installations or dedicated server infrastructure.

This design choice makes the system more accessible and portable across common hardware platforms. It also supports easier testing and deployment, since users can access the application through a browser. However, client-side execution may also introduce limitations related to browser performance, memory usage, file size, and hardware capability. These limitations should be considered in future testing, particularly when handling longer recordings or larger IMU datasets.

5.2 Sensor Fusion Algorithm Comparison

The three sensor-fusion pipelines demonstrated substantially different behaviour across the synthetic validation datasets. The following sections compare their performance under rotational and mixed-motion conditions, with particular focus on the mechanisms responsible for the observed differences.

5.2.1 Suitability of Gradient-Descent Fusion Under Dynamic Rotation

The Expanded Madgwick filter performed unacceptably for every dataset involving rotation, accumulating yaw errors of 430–1293° against ground-truth values of 90–180°. Similarly, the peak linear acceleration residual during Dataset 2's pure-rotation sequence was approximately 90× greater for Madgwick than for the two successful algorithms (19.75 vs 0.22 $\mathrm{m\,s^{-2}}$). This

rules out incorrect parameter tuning as the primary cause of failure, since the filter demonstrated acceptable behaviour at rest and diverged only during rotational motion. It should also be noted that the Expanded Madgwick pipeline benefited from a retroactive linear drift correction applied between ZUPT pairs, a post-processing step that was not applied to VQF or EKF. Its relatively competitive distance estimate on Dataset 1 (3 %) may therefore partially reflect this additional correction rather than superior kinematic integration.

The root cause of this issue is a fundamental ambiguity in the gradient-descent formulation. During simultaneous pitch and roll rotation, the projection of the gravity vector onto the sensor axes changes continuously. This is indistinguishable from translational acceleration in the Madgwick cost function, producing substantial orientation and residual errors during sustained multi-axis rotation. The binary accelerometer gate can mitigate this ambiguity during discrete translational bursts but fails to suppress it during sustained multi-axis rotation, where the gate criterion (specific-force departure from $1\,g$ by more than $0.1\,g$) is not reliably satisfied.

This result is consistent with the known limitations of gradient-descent methods and indicates that the Expanded Madgwick filter is unsuitable for any application scenario involving concurrent rotation, including the turning and rolling manoeuvres present in most athletic activities. Consequently, subsequent sections adopt an alternative sensor-fusion approach.

5.2.2 Continuous versus Binary Accelerometer Correction

Both the VQF and EKF pipelines achieved accurate orientation estimates in the linear and rotational datasets, but diverged on Dataset 3 (mixed motion). EKF's binary gate achieves near-perfect yaw recovery on Dataset 2 ($+0.4^\circ$ error) by completely suppressing accelerometer input during dynamic rotation. However, this same strategy introduces a larger kinematic deficit on Dataset 3, where EKF exhibits greater errors than VQF in both distance ($-14\,\%$ vs $-11\,\%$) and peak speed ($-20\,\%$ vs $-10\,\%$), alongside a roll error of 13 % versus 1 %. The likely cause of these errors is that simultaneous linear acceleration and rotation renders the gate criterion unreliable. In this case, the sensed specific force departs from $1\,g$ in a direction determined by both the translational acceleration and the changing gravity projection, causing the gate to open and close in a way that is not strictly aligned with translational events. Conversely, VQF's continuous low-pass correction avoids these abrupt gate transitions. Instead it attenuates high-frequency orientation updates slightly, without exposing the kinematic estimate to classification errors that arise when translation and rotation overlap.

The performance trade-off reverses on Dataset 2, where VQF produces a $+9.2^\circ$ yaw surplus that EKF avoids entirely. This was consistent with the τ_{acc} time constant reducing the responsiveness of tilt correction during dynamic reorientation, allowing small orientation errors to persist into the heading estimate. For applications involving combined translational and rotational motion, as encountered in real athletic activity, VQF's advantage on Dataset 3 outweighs EKF's advantage on Dataset 2, justifying its selection as the basis for the parameter sensitivity study.

5.3 VQF Parameter Sensitivity

The parameter sensitivity study examined how the VQF accelerometer and magnetometer time constants influenced orientation and kinematic performance across all synthetic datasets. The following sections discuss the structural behaviour of these parameters and the practical implications of the observed trends.

5.3.1 Parameter Orthogonality

The complete independence of τ_{acc} and τ_{mag} across all 147 parameter combinations was a practically significant structural result. Because the accelerometer correction acts exclusively on the

tilt axes (pitch and roll) while the magnetometer correction acts exclusively on the heading axis (yaw), the two parameters control entirely non-overlapping aspects of filter output. For an end user, this means kinematic accuracy and heading accuracy can be tuned sequentially. τ_{acc} can be set by minimising distance or speed error on a known linear motion, then τ_{mag} can be set independently by minimising yaw error on a known rotation, with no need to iterate between the two.

5.3.2 Parameter Sensitivity and Operating Range

The uniform improvement with increasing τ_{acc} across both the linear and mixed datasets confirmed that the optimal strategy under the synthetic data noise model was to maximise gyroscope trust during dynamic phases. The observed gains did exhibit clear diminishing returns above $\tau_{\text{acc}} = 10\,\text{s}$, where the improvement from 10 to 15 s was +0.21 m on mixed distance and fell to +0.11 m from 15 to 20 s. A further consideration was that when $\tau_{\text{acc}} = 20\,\text{s}$ the correction time constant exceeds the duration of all the synthetic recordings. The accelerometer thus contributes negligibly to tilt estimation throughout the session, and the filter operates close to pure gyroscope integration. While this is numerically optimal on 18-second synthetic recordings with a clean, bias-corrected gyroscope, it reflects a limiting case of the filter behaviour rather than a generalised tuning recommendation. The marginal gain from 15 to 20 s is an artefact of recording length, not filter capability. $\tau_{\text{acc}} = 15\,\text{s}$ was therefore selected as the practically optimal value to continue with physical experimentation and was set as the application default. It captured the majority of the available kinematic improvement (+0.72 m on mixed distance relative to the default) while retaining meaningful accelerometer influence and remaining within the recording duration.

In real IMU recordings, very large τ_{acc} values increase susceptibility to uncorrected gyroscope bias drift between ZUPT events. This arises because the accelerometer has reduced influence to correct slow drift accumulation in tilt estimation. The synthetic noise model uses a Gaussian bias of fixed magnitude, which is fully corrected at initialisation, whereas real MEMS gyroscopes exhibit time-varying bias drift that may not be fully eliminated by a one-time static calibration. The optimal τ_{acc} for real hardware recordings may therefore be lower than the value identified here, reflecting a balance between translational acceleration rejection and bias drift correction that cannot be characterised from synthetic data alone.

For τ_{mag}, the improvement curve flattened sharply above 30 s. The step from $\tau_{\text{mag}} = 60\,\text{s}$ to 90 s yielded only 0.5° reduction in rotation yaw error. Because this improvement is indistinguishable from gyroscope noise and 60 s is not the tested boundary, $\tau_{\text{mag}} = 60\,\text{s}$ was selected as the practically optimal value for future experimentation and the web app default configuration. At $\tau_{\text{mag}} = 90\,\text{s}$ over an 18-second recording the magnetometer correction is similarly negligible, and the same limitation of long time constants reducing responsiveness to magnetic disturbances, including soft iron effects and nearby ferromagnetic structures, applies. Ultimately, the optimal value must be assessed based on the real recording environment rather than synthetic data alone.

5.4 Limitations of Synthetic Validation

The synthetic evaluation provides a controlled basis for algorithm selection and parameter characterisation, but several limitations constrain direct extrapolation to real-world performance. First, the synthetic noise model was strictly Gaussian and temporally uncorrelated. Real MEMS sensors can exhibit heavier-tailed distributions, correlated noise, and vibration-induced artefacts that are not captured in the synthetic data. Second, the synthetic trajectories are analytically smooth. The acceleration profiles used trapezoidal ramps with no jerk transients, and the rotation sweeps were executed at constant angular rates. Real motion involves continuous, overlapping acceleration and rotation components that more strongly challenge the fusion algorithms. Third, no magnetic disturbance was modelled in the synthetic data, unlike real recordings. The

magnetometer provided a clean heading reference under the synthetic assumptions, favouring both VQF and EKF equally. The real-world experiments described below address these limitations under controlled experimental conditions.

5.5 Real-World Validation

The real-world experiments were used to evaluate whether the behaviour observed during synthetic validation remained consistent under physical recording conditions. The following sections discuss the outcomes of the attempted linear-motion validation together with the controlled rotation and mixed-motion experiments.

5.5.1 Linear Motion Experiment

Validation of the pipeline's distance estimation capability against known linear displacement proved considerably more challenging than the synthetic evaluation suggested. Four physical approaches were investigated: a wheeled suitcase, table-surface dragging, free hand-carry through air, and a drawer-mounted setup.

Across all cases, the dominant error source was not integration performance but unavoidable rotational contamination introduced by the experimental platforms. The wheeled suitcase transmitted angular vibrations through its chassis despite reduced rolling friction, producing large orientation excursions and inconsistent trajectories across repeated trials. Table-surface dragging removed much of this drift but exposed a different limitation. At low speeds, true translational acceleration (approximately $0.05\,g$) was comparable to gravity leakage caused by small orientation errors, making stable velocity integration unachievable even with ZUPT correction. Increasing speed improved signal separation but reintroduced rotational vibration from frictional effects, producing a fundamental speed–stability trade-off. Free hand-carry motion in air produced large angular velocities, which dominated the gyroscope signal and corrupted gravity compensation throughout. The drawer-mounted configuration introduced additional magnetic disturbances from nearby metallic components, further degrading the heading stability.

None of the tested configurations produced reliable or consistent distance estimates, with all trials significantly under- or over-estimating ground truth. As a result, all four approaches were deemed unsuitable for quantitative validation of linear displacement. This outcome indicates that inertial-only distance estimation is highly sensitive to rotational contamination in unconstrained environments, and that meaningful validation requires some form of mechanically constrained motion (e.g., rail or linear track).

5.5.2 Rotation Experiment

Yaw Overshoot All three trials produced a mean yaw of approximately 94.7° against a ground truth of 90°, a consistent overshoot of +5.2 %. This appears to be a systematic error rather than a random error and is thus unlikely to arise from noise alone. The most probable cause is a small constant gyroscope bias from the IMU on the yaw axis that survived the calibration process. The calibration window estimates bias from a brief static hold, and any residual bias integrates linearly with rotation duration, producing a proportional overestimate. The consistency of sign and magnitude across all trials supports a fixed-bias explanation over the alternative contribution of the finger-pivot constraint, which would be expected to vary more between trials. The inter-trial variation in yaw error (+3.4 % to +7.5 %) is consistent with the expected level of noise across short recordings and does not suggest a systematic per-trial cause.

Roll Asymmetry Between Spins The first spin from each trial produced a mean roll range of 5.75° whereas the second spin produced 10.80°. Because the IMU ends the first spin at approximately 90° from its starting orientation, the sensor axes have rotated relative to the

world frame. In the second spin the axis that gravity acts on during the rotation differs from the first, causing the gravity-removal step to interact differently with any residual gravity-leakage on the roll channel. The asymmetry identified here therefore appears to be an artefact of the orientation-dependent coupling between gravity projection and the roll estimate rather than a change in physical motion between the two spins.

Pitch Range The pitch ranges were small (mean 3.92°) and consistent across trials. The IMU was placed on a flat surface and rotated in-plane, so true pitch displacement was zero. The observed pitch range is attributed to minor surface irregularities causing the IMU to rock slightly during the pivot, and to cross-axis coupling from the same gravity-leakage mechanism described above. The consistency across trials indicates this is also a systematic floor on pitch accuracy for flat in-plane rotation rather than random noise.

Magnetometer Interference and Recording Duration The magnetometer readings showed distortion consistent with residual soft-iron effects and incomplete hard-iron compensation from nearby ferromagnetic objects. To limit any heading drift accumulation, each recording was kept short (approximately 11 seconds total), and the two 90° rotations were analysed as separate movements rather than as a single continuous recording. Analysing each spin independently confined any heading drift to within one range window, preventing accumulation across the full recording. This approach produced stable yaw estimates within each range. However, a longer or multi-step rotation protocol may require either a magnetically clean environment or a magnetometer-free pipeline configuration.

5.5.3 Mixed Motion Experiment

For the mixed motion experiment the distance estimation failed significantly, with a mean estimated distance of 4.18 m against a ground truth of 0.71 m, an overestimation of +489 %. The IMU was mounted on the flat end face of the jar and underwent no tumbling motion during rolling, meaning its trajectory was purely translational with a simultaneous yaw rotation. The distance failure therefore has the same underlying cause as in the linear experiment. The double integration of small translational accelerations accumulated drift rapidly, and residual gravity leakage from orientation error dominated the true motion signal. Despite this, the yaw tracking remained accurate throughout with a mean error of +0.82 %, confirming that the orientation pipeline itself performed correctly and that the distance error is attributable entirely to kinematic integration limitations rather than failure of the orientation estimate.

5.6 Choice of Metrics

The selected metrics provide a balanced and broadly applicable representation of athlete movement, capturing both how hard the athlete is working and how that effort is distributed over time, without relying on overly sport-specific features. Together, they cover key performance indicators seen across multiple sports such as soccer, golf, strength training, and injury assessment.

Another reason for including these metrics is that they are relatively intuitive and easy for a regular user to understand. In addition, their selection reflects a trade-off between excessive complexity and oversimplification. Including too many metrics would introduce redundancy, increase complexity, and make interpretation more difficult, while also leading to overlapping information across highly correlated measures. In contrast, selecting too few metrics would oversimplify movement and limit the range of meaningful performance applications.

The set of metrics was also designed to minimise overlap, ensuring that each variable contributes distinct and complementary information. In this way, each metric captures a relatively

unique aspect of movement, resulting in a feature set that is informative without being redundant.

5.7 Ethics

Establishing an ethical framework is necessary to ensure user safety, safeguard personal data, and maintain public trust in digital systems. As this project involves the processing of human motion data derived from video recordings and IMU sensors, it introduces several ethical aspects that must be addressed. These include privacy and data protection, informed consent, data management practices, potential misinterpretation of results, and the use of third-party and open-source software. The considerations outlined here extend the discussion from the previous PCC1 and PCC2 reports, as this work continues the development of the same system.

5.7.1 Privacy and Data Protection

Privacy and responsible handling of data are key ethical priorities in this project, especially since video recordings may include identifiable personal information. To reduce these risks, the system is designed to operate entirely within the user's web browser. As a result, neither video data nor derived motion information is transmitted to external servers or cloud-based services; all processing takes place locally on the user's device.

Any project-related metadata is stored locally in the browser environment, with the option for users to export or import it as needed. This ensures that users maintain full ownership and control over their data throughout the entire process. This approach follows the principle of data minimisation and significantly reduces exposure to risks such as unauthorized access or data breaches, thereby supporting user trust.

5.7.2 Informed Consent

Informed consent, which is closely tied to the principle of respect for autonomy within bioethical principlism, is another essential consideration. It refers to the right of individuals to make voluntary and well-informed decisions regarding their participation after receiving adequate information about the system and its use.

During the development phase, all testing was conducted on a voluntary basis and solely for educational and prototyping purposes. No external participants were involved, as the system was tested exclusively by members of the project group. Should the system be further developed for use in real-world contexts such as clinical rehabilitation or sports analysis, formal consent procedures and clear information would be required to ensure that users fully understand how their data is collected and processed.

5.7.3 Misinterpretation and Over-reliance

There is also a risk that system outputs may be misinterpreted or that users may rely too heavily on the results. Given that the motion metrics are limited in accuracy and have not undergone clinical validation the system is not suitable for diagnostic or treatment-related decisions. Instead, it should be used as an exploratory or supportive tool for analysing movement patterns.

This limitation also relates to accessibility and responsible use. While the browser-based and low-cost design improves accessibility compared to traditional motion capture systems, it may also lead to use by individuals who lack the expertise to correctly interpret its limitations. For this reason, appropriate disclaimers and usage guidelines would be necessary in any deployed version of the system to ensure that users understand its intended scope and constraints.

5.7.4 Third-party and Open-source code

Ethical considerations also arise from the use of third-party and open-source software. A substantial part of the system's implementation is built upon publicly available code and resources from platforms such as GitHub and other technical documentation sources. This choice was made in accordance with the project supervisor's recommendation and reflects common practice in modern software development, particularly in academic settings where efficiency and transparency are important.

All external code was carefully reviewed, adapted, and integrated to meet the requirements of the system. In parallel, the code developed within this project has been released publicly via a GitHub repository. This decision promotes transparency, reproducibility, and opportunities for further improvement by others. By contributing the code back to the open-source community, the project participates in and supports the same ecosystem from which it benefited during development.

5.8 Future Work

Although the implemented system achieved the primary objectives of this project, several limitations and unresolved challenges remain. The following sections outline areas for further development related to usability, synchronization, sensor fusion, validation, and system robustness.

5.8.1 Evaluation of the Web Application

Future work should focus on moving MoveSync from a functional prototype toward a validated and more robust motion-analysis system. The current implementation provides a coherent foundation for managing, inspecting, processing, and comparing motion-tracking sessions. However, the results mainly demonstrate functional implementation rather than full validation of measurement accuracy or practical usability.

The web application should therefore be further evaluated with intended users to assess navigation, workflow clarity, and ease of use during complete analysis sessions. This would help identify usability issues that are not visible from interface implementation alone, particularly in the *Upload*, *Library*, *Session Viewer*, and *Compare Sessions* modules. User testing would also make it possible to evaluate whether the modular workflow supports practical movement analysis clearly and efficiently.

5.8.2 Improved Synchronization Workflow

The synchronization workflow should also be improved. In the current implementation, the time synchronization still depends on manual event selection, which may introduce user-dependent variability. Future versions should therefore include more automated synchronization procedures, for example by detecting characteristic motion events in the IMU signals and aligning them with corresponding events in the video. This would reduce manual interaction and improve repeatability between users and sessions.

Furthermore, the current synchronization approach is mainly based on the assumption that the dominant discrepancy between the video and IMU data streams is a fixed start-time offset. This assumption should be evaluated further using real motion recordings. For longer recordings or recordings with unstable timing behavior, additional synchronization reference points or more advanced alignment models may be required to account for effects such as clock drift, dropped frames, or inconsistent sampling intervals.

5.8.3 Further Development of the Session Viewer

Further development of the *Session Viewer* should focus on expanding and validating the analysis tools. The MoveNet-based angle analysis could be extended with automatic selection of commonly used joint angles, storage of angle-time curves, and export of angle metrics. The Sensor Fusion 3D view should also be refined so that the orientation visualization can be compared more directly with the video and IMU graphs.

In addition, the expanded analysis tab could be developed further to present a clearer summary of session-level metrics and their interpretation. This would make the Session Viewer more useful not only for inspecting individual signals but also for summarizing the main outcomes of a recording in a structured way.

5.8.4 Extension of the Compare Sessions Module

The *Compare Sessions* module should be extended to support more detailed quantitative comparison between recordings. Future work could include automatic normalization of movement phases, clearer visualization of metric differences, and support for comparing more than two sessions. This would make the module more useful for repeated trials, longitudinal tracking, and comparison between subjects or movement conditions.

The interpretation of session comparisons should also be evaluated under controlled conditions. Differences between sessions may be caused by actual movement differences, but they may also be influenced by synchronization accuracy, sensor placement, recording conditions, and preprocessing settings. Future testing should therefore examine how sensitive the comparison results are to these factors.

5.8.5 Validation of the Sensor-Fusion Pipeline

The sensor-fusion pipeline requires further validation using real IMU recordings. Although the VQF-based approach was selected based on the synthetic evaluation and subsequent experiments, the optimal parameter settings should be tested on a larger range of real movements and recording conditions. Particular attention should be given to gyroscope bias, sensor placement, calibration quality, and magnetic disturbances, since these factors strongly affect orientation estimation and downstream analysis.

Future work should also investigate how different preprocessing configurations influence the reliability and repeatability of the displayed results. This includes evaluating the effects of filtering, calibration, zero-velocity update settings, and other processing parameters on IMU-derived signals, extracted metrics, and Sensor Fusion 3D visualization.

5.8.6 Improved Validation of Distance Estimation

Distance estimation remains a major limitation and should be investigated using a more controlled validation setup. Future experiments should use mechanically constrained motion, such as a rail, linear track, or encoder-based platform, to provide repeatable translational motion with minimal rotational contamination. The results should also be compared against an external reference system, such as optical motion capture or another validated tracking method, to quantify accuracy more reliably.

5.8.7 Broader Testing and Generalization

Future work should include broader testing across different sports, rehabilitation tasks, and movement types. This would make it possible to evaluate whether the developed metrics, presets, and visualization tools generalize beyond the initial test cases. Testing with a wider range of

movements would also help determine whether the application workflow remains usable when applied to different analysis scenarios.

With improved validation, automated synchronization, refined preprocessing procedures, and further development of the comparison and reporting functions, MoveSync could be developed into a more reliable tool for practical motion analysis.

6 Conclusion

This project aimed to develop a low-cost, accessible, browser-based prototype for sports motion analysis integrating video pose estimation and IMU sensing. The resulting system, MoveSync, achieves the three main objectives. Browser-based skeletal keypoint extraction via MoveNet and TensorFlow.js is implemented and functional, a manual event-based synchronisation method successfully aligns the independent video and IMU time bases, and a structured multi-module interface provides synchronised visualisation and inspection of the combined data streams. The system operates entirely within the web application, requiring no downloads or dedicated server infrastructure, which directly supports the accessibility goals of the project.

The IMU signal processing pipeline was substantially advanced through a systematic comparison of three sensor fusion algorithms. The VQF algorithm was identified as the most appropriate approach for this application, offering robust and accurate orientation estimation under mixed translational and rotational motion. The parameter sensitivity study demonstrated that the two VQF time constants are fully orthogonal and can be tuned independently, a practically useful result that simplifies calibration for end users. The optimised configuration ($\tau_{\text{acc}} = 15\,\text{s}$, $\tau_{\text{mag}} = 60\,\text{s}$) was validated in physical experiments, achieving a mean yaw tracking error of $+5.2\,\%$ across repeated $90°$ rotation steps, and configured as the default within the web application.

The primary limitation identified in this project is inertial distance estimation under low-acceleration conditions. Both synthetic evaluation and physical experiments demonstrated that double integration of MEMS accelerometer data accumulates drift rapidly in unconstrained environments, making reliable distance quantification infeasible without mechanical constraint or external correction. This represents the most significant barrier to expanding the system's quantitative capability and is the clearest direction for future development. Orientation tracking, by contrast, performed well and can be considered a validated capability of the current pipeline.

MoveSync demonstrates that a functional, multimodal motion analysis system can be built and deployed within a browser without specialised hardware or server infrastructure. While the current implementation is best regarded as a validated prototype rather than a production tool, the modular architecture, principled sensor fusion pipeline, and identified improvement priorities provide a strong foundation for continued development toward practical use in real-world sports analysis contexts.

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Appendix A



Figure A1: The IMU sensor used for all experimental recording and testing, the Movesense model OP174, usually connected to the Movesense Showcase mobile application.



Figure A2: The experimental setups. The mixed motion experiment (left) features the IMU attached to the end of a jar. The rotational experiment (right) has the IMU placed flat on a marked surface to measure rotation.